Main Category	Category	Terms
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Apoptosis
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Cell cycle
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Transcription
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Transcription regulation
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Host-virus interaction
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Nucleus
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Activator
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	DNA-binding
Functional Annotations	UP_KW_PTM	Acetylation
Functional Annotations	UP_KW_PTM	Methylation
Functional Annotations	UP_KW_PTM	Phosphoprotein
Functional Annotations	UP_KW_PTM	Ubl conjugation
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Low complexity
Functional Annotations	UP_SEQ_FEATURE	MOTIF:DEF box
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Abolishes acetylation; when associated with R-117 and R-120.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Abolishes acetylation; when associated with R-117 and R-125.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Abolishes acetylation; when associated with R-120 and R-125.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Abrogates methylation by SETD7. Loss of interaction with L3MBTL3. Loss of ubiquitination by the CRL4-DCAF5 E3 ubiquitin ligase complex.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:L->E: Abolishes interaction with and repression of CEBPA and inhibition of adipogenesis.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:S->A: Abrogates in vitro phosphorylation by CHEK2 and CHEK2-dependent stabilization of E2F1 upon DNA damage.

Main Category	Category	Terms
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:S->A: Decreased phosphorylation by GSK3B, leading to abolished interaction with USP11 and subsequent deubiquitination.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:T->A: Decreased phosphorylation by GSK3B.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:Y->C: No retinoblastoma protein binding. No effect on interaction with and repression of CEBPA.
Functional Annotations	UP_SEQ_FEATURE	REGION:Cyclin A:CDK2 binding
Functional Annotations	UP_SEQ_FEATURE	REGION:Dimerization
Functional Annotations	UP_SEQ_FEATURE	REGION:Disordered
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with BIRC2/c-IAP1
Functional Annotations	UP_SEQ_FEATURE	REGION:Leucine-zipper
Functional Annotations	UP_SEQ_FEATURE	REGION:RB1 binding
Functional Annotations	UP_SEQ_FEATURE	REGION:Required for interaction with TRIM28
Functional Annotations	UP_SEQ_FEATURE	REGION:Transactivation
Gene Ontology	GOTERM_BP_DIRECT	DNA damage checkpoint signaling
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	DNA-templated transcription
Gene Ontology	GOTERM_BP_DIRECT	regulation of DNA-templated transcription
Gene Ontology	GOTERM_BP_DIRECT	regulation of transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	spermatogenesis
Gene Ontology	GOTERM_BP_DIRECT	intrinsic apoptotic signaling pathway in response to DNA damage
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of gene expression
Gene Ontology	GOTERM_BP_DIRECT	forebrain development

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	response to lipopolysaccharide
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	anoikis
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of DNA binding
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of fat cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of DNA-templated transcription
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of DNA-templated transcription
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of fibroblast proliferation
Gene Ontology	GOTERM_BP_DIRECT	mRNA stabilization
Gene Ontology	GOTERM_BP_DIRECT	regulation of cell cycle
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of glial cell proliferation
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of fat cell proliferation
Gene Ontology	GOTERM_BP_DIRECT	cellular response to fatty acid
Gene Ontology	GOTERM_BP_DIRECT	cellular response to hypoxia
Gene Ontology	GOTERM_BP_DIRECT	cellular response to xenobiotic stimulus
Gene Ontology	GOTERM_BP_DIRECT	intrinsic apoptotic signaling pathway by p53 class mediator
Gene Ontology	GOTERM_BP_DIRECT	lens fiber cell apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	cellular response to nerve growth factor stimulus
Gene Ontology	GOTERM_BP_DIRECT	regulation of G1/S transition of mitotic cell cycle
Gene Ontology	GOTERM_CC_DIRECT	nuclear chromosome

Main Category	Category	Terms
Gene Ontology	GOTERM_CC_DIRECT	chromatin
Gene Ontology	GOTERM_CC_DIRECT	nucleus
Gene Ontology	GOTERM_CC_DIRECT	nucleoplasm
Gene Ontology	GOTERM_CC_DIRECT	transcription regulator complex
Gene Ontology	GOTERM_CC_DIRECT	cytoplasm
Gene Ontology	GOTERM_CC_DIRECT	centrosome
Gene Ontology	GOTERM_CC_DIRECT	protein-containing complex
Gene Ontology	GOTERM_CC_DIRECT	Rb-E2F complex
Gene Ontology	GOTERM_CC_DIRECT	RNA polymerase II transcription regulator complex
Gene Ontology	GOTERM_MF_DIRECT	transcription cis-regulatory region binding
Gene Ontology	GOTERM_MF_DIRECT	RNA polymerase II cis-regulatory region sequence-specific DNA binding
Gene Ontology	GOTERM_MF_DIRECT	DNA-binding transcription factor activity, RNA polymerase II-specific
Gene Ontology	GOTERM_MF_DIRECT	cis-regulatory region sequence-specific DNA binding
Gene Ontology	GOTERM_MF_DIRECT	DNA-binding transcription activator activity
Gene Ontology	GOTERM_MF_DIRECT	DNA binding
Gene Ontology	GOTERM_MF_DIRECT	DNA-binding transcription factor activity
Gene Ontology	GOTERM_MF_DIRECT	protein binding
Gene Ontology	GOTERM_MF_DIRECT	protein kinase binding
Gene Ontology	GOTERM_MF_DIRECT	sequence-specific DNA binding
Gene Ontology	GOTERM_MF_DIRECT	protein dimerization activity
Gene Ontology	GOTERM_MF_DIRECT	molecular adaptor activity

Main Category	Category	Terms
Gene Ontology	GOTERM_MF_DIRECT	DNA-binding transcription factor binding
Gene Ontology	GOTERM_MF_DIRECT	sequence-specific double-stranded DNA binding
General Annotations	CHROMOSOME	20
General Annotations	CYTOGENETIC_LOCATION	20q11.22
General Annotations	ENTREZ_GENE_SUMMARY	The protein encoded by this gene is a member of the E2F family of transcription factors. The E2F family plays a crucial role in the control of cell cycle and action of tumor suppressor proteins and is also a target of the transforming proteins of small DNA tumor viruses. The E2F proteins contain several evolutionally conserved domains found in most members of the family. These domains include a DNA binding domain, a dimerization domain which determines interaction with the differentiation regulated transcription factor proteins (DP), a transactivation domain enriched in acidic amino acids, and a tumor suppressor protein association domain which is embedded within the transactivation domain. This protein and another 2 members, E2F2 and E2F3, have an additional cyclin binding domain. This protein binds preferentially to retinoblastoma protein pRB in a cell-cycle dependent manner. It can mediate both cell proliferation and p53-dependent/ independent apoptosis. [provided by RefSeq, Jul 2008]
General Annotations	UP_COMMENT_ACTIVITY_REGULATION	BIRC2/c-IAP1 stimulates its transcriptional activity. {ECO:0000269|PubMed:21653699}.
General Annotations	UP_COMMENT_FUNCTION	Transcription activator that binds DNA cooperatively with DP proteins through the E2 recognition site, 5'-TTTC[CG]CGC-3' found in the promoter region of a number of genes whose products are involved in cell cycle regulation or in DNA replication (PubMed:10675335, PubMed:12717439, PubMed:17050006, PubMed:17704056, PubMed:18625225, PubMed:28992046). The DRTF1/ E2F complex functions in the control of cell-cycle progression from G1 to S phase (PubMed:10675335, PubMed:12717439, PubMed:17704056). E2F1 binds preferentially RB1 in a cell-cycle dependent manner (PubMed:10675335, PubMed:12717439, PubMed:17704056). It can mediate both cell proliferation and TP53/p53- dependent apoptosis (PubMed:8170954). Blocks adipocyte differentiation by binding to specific promoters repressing CEBPA binding to its target gene promoters (PubMed:20176812). Directly activates transcription of PEG10 (PubMed:17050006, PubMed:18625225, PubMed:28992046). Positively regulates transcription of RRP1B (PubMed:20040599). {ECO:0000269| PubMed:10675335, ECO:0000269|PubMed:12717439, ECO:0000269|PubMed:17050006, ECO:0000269| PubMed:17704056, ECO:0000269|PubMed:18625225, ECO:0000269|PubMed:20040599, ECO:0000269|

Main Category	Category	Terms
		PubMed:20176812, ECO:0000269|PubMed:28992046, ECO:0000269|PubMed:8170954}.
General Annotations	UP_COMMENT_INTERACTION	Q01094; P05067: APP; NbExp=3; IntAct=EBI-448924, EBI-77613; Q01094; Q14201: BTG3; NbExp=3; IntAct=EBI-448924, EBI-948192; Q01094; P49336: CDK8; NbExp=3; IntAct=EBI-448924, EBI-394377; Q01094; Q92466: DDB2; NbExp=2; IntAct=EBI-448924, EBI-1176171; Q01094; Q13547: HDAC1; NbExp=2; IntAct=EBI-448924, EBI-301834; Q01094; Q92830: KAT2A; NbExp=3; IntAct=EBI-448924, EBI-477622; Q01094; Q96JM7: L3MBTL3; NbExp=2; IntAct=EBI-448924, EBI-2686809; Q01094; Q8NEM0: MCPH1; NbExp=6; IntAct=EBI-448924, EBI-1565483; Q01094; Q9Y618: NCOR2; NbExp=2; IntAct=EBI-448924, EBI-80830; Q01094; P09874: PARP1; NbExp=3; IntAct=EBI-448924, EBI-355676; Q01094; Q96IZ0: PAWR; NbExp=2; IntAct=EBI-448924, EBI-595869; Q01094; O14744: PRMT5; NbExp=8; IntAct=EBI-448924, EBI-351098; Q01094; P06400: RB1; NbExp=29; IntAct=EBI-448924, EBI-491274; Q01094; Q14684: RRP1B; NbExp=10; IntAct=EBI-448924, EBI-372051; Q01094; Q96EB6: SIRT1; NbExp=3; IntAct=EBI-448924, EBI-1802965; Q01094; P08047: SP1; NbExp=2; IntAct=EBI-448924, EBI-298336; Q01094; P42224: STAT1; NbExp=2; IntAct=EBI-448924, EBI-1057697; Q01094; Q14186: TFDP1; NbExp=17; IntAct=EBI-448924, EBI-749713; Q01094; Q92547: TOPBP1; NbExp=3; IntAct=EBI-448924, EBI-308302; Q01094; O95361: TRIM16; NbExp=2; IntAct=EBI-448924, EBI-727384; Q01094; P03129: E7; Xeno; NbExp=2; IntAct=EBI-448924, EBI-866453; Q01094; Q923E4: Sirt1; Xeno; NbExp=3; IntAct=EBI-448924, EBI-1802585;
General Annotations	UP_COMMENT_PTM	Acetylation stimulates DNA-binding. Enhanced under stress conditions such as DNA damage and inhibited by retinoblastoma protein RB1. Regulated by KAP1/TRIM28 which recruits HDAC1 to E2F1 resulting in deacetylation. Acetylated by P/CAF/KAT2B. {ECO:0000269|PubMed:10675335, ECO:0000269|PubMed:17704056}.
General Annotations	UP_COMMENT_PTM	Methylation at Lys-185 by SETD7 promotes E2F1 ubiquitin-dependent proteasomal degradation. {ECO:0000269|PubMed:20603083, ECO:0000269|PubMed:29691401}.
General Annotations	UP_COMMENT_PTM	Phosphorylated by CDK2 and cyclin A-CDK2 in the S-phase (PubMed:12717439, PubMed:7838523). Phosphorylation at Ser-364 by CHEK2 stabilizes E2F1 upon DNA damage and regulates its effect on transcription and apoptosis (PubMed:12717439). Phosphorylation at Ser- 403 by GSK3B promotes interaction with USP11, leading to its deubiquitination and stabilization (PubMed:28992046). {ECO:0000269|PubMed:12717439, ECO:0000269|PubMed:28992046, ECO:0000269|PubMed:7838523}.
General Annotations	UP_COMMENT_PTM	Ubiquitinated via 'Lys-63'-linked ubiquitin, leading to its degradation (PubMed:28992046). Deubiquitinated by

Main Category	Category	Terms
		USP11 following phosphorylation by GSK3B, promoting its stability (PubMed:28992046). {ECO:0000269|PubMed:28992046}.
General Annotations	UP_COMMENT_SEQUENCE_CAUTION	Sequence=AAB24289.1; Type=Erroneous initiation; Note=Extended N-terminus.; Evidence={ECO:0000305};
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the E2F/DP family. {ECO:0000305}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Nucleus {ECO:0000269|PubMed:20040599, ECO:0000269|PubMed:28992046}.
General Annotations	UP_COMMENT_SUBUNIT	(Microbial infection) Interacts with human cytomegalovirus/HHV-5 protein UL123. {ECO:0000269|PubMed:7494286}.
General Annotations	UP_COMMENT_SUBUNIT	Component of the DRTF1/E2F transcription factor complex. Forms heterodimers with DP family members. The E2F1 complex binds specifically hypophosphorylated RB1, the interaction represses E2F1- driven transcription (PubMed:8336704). During the cell cycle, RB1 becomes phosphorylated in mid-to-late G1 phase, detaches from the DRTF1/E2F complex, rendering E2F transcriptionally active. Viral oncoproteins, notably E1A, T-antigen and HPV E7, are capable of sequestering RB1, thus releasing the active complex. Interacts with TRRAP, which probably mediates its interaction with histone acetyltransferase complexes, leading to transcription activation. Binds TOPBP1 and EAPP. Interacts with ARID3A. Interacts with TRIM28; the interaction inhibits E2F1 acetylation through recruiting HDAC1 and represses its transcriptional activity. Interaction with KAT2B; the interaction acetylates E2F1 enhancing its DNA-binding and transcriptional activity. Interacts with BIRC2/c-IAP1 (via BIR domains). The complex TFDP1:E2F1 interacts with CEBPA; the interaction prevents CEBPA binding to target genes promoters and represses its transcriptional activity (PubMed:20176812). Interacts with RRP1B (PubMed:20040599). Interacts with HCFC1 (PubMed:23629655). Interacts with KMT2E; the interaction is probably indirect and is mediated via HCFC1 (PubMed:23629655). Interacts with DCAF5 and L3MBTL3; the interaction requires methylation at Lys-185 and is necessary to target E2F1 for ubiquitination by the CRL4-DCAF5 E3 ubiquitin ligase complex (PubMed:29691401). {ECO:0000250|UniProtKB:Q61501, ECO:0000269|PubMed:10675335, ECO:0000269|PubMed:11418595, ECO:0000269|PubMed:12598654, ECO:0000269|PubMed:12697828, ECO:0000269|PubMed:15716352, ECO:0000269|PubMed:16360038, ECO:0000269|PubMed:17704056, ECO:0000269|PubMed:20040599, ECO:0000269|PubMed:20176812, ECO:0000269|PubMed:21653699, ECO:0000269|PubMed:23629655, ECO:0000269|PubMed:29691401, ECO:0000269|PubMed:8336704, ECO:0000269|PubMed:9780002}.
Identifiers	ENTREZ_GENE_ID	1869

Main Category	Category	Terms
Identifiers	ENSEMBL_GENE_ID	ENSG00000101412
Identifiers	UNIPROT_ID	Q9BSD8_HUMAN
Identifiers	UNIPROT_ID	E2F1_HUMAN
Literature	GENERIF_SUMMARY	transcriptional inhibition of the plasminogen activator inhibitor type 1 gene
Literature	GENERIF_SUMMARY	expression reduced in primary and metastatic breast carcinoma
Literature	GENERIF_SUMMARY	selection and characterization of active hammerhead ribozymes targeted against E2F1 full-length mRNA
Literature	GENERIF_SUMMARY	Regulation of E2F1-dependent gene transcription and apoptosis by the ETS-related transcription factor GABPgamma1
Literature	GENERIF_SUMMARY	increased ppRb and E2FI immunoreactivity in Alzheimer disease brain, with ppRb predominately located in the nucleus and E2F1 in the cytoplasm.
Literature	GENERIF_SUMMARY	Human telomerase accelerates growth of lens epithelial cells through regulation of the genes mediating RB/E2F pathway.
Literature	GENERIF_SUMMARY	Overexpression of E2F-1 induces apoptosis and increases chemosensitivity in human pancreatic carcinoma cells. Cell cycle, PARP cleavage and morphology supported apoptosis as the cell death mechanism in reponse to E2F1.
Literature	GENERIF_SUMMARY	conclude that E2F proteins and Sp1 play an important role in the control of p18 expression
Literature	GENERIF_SUMMARY	p14(ARF) regulates E2F activity in different cell types and down-regulates E2F-dependent transcription, and in cells undergoing E2F-dependent apoptosis prompts cell cycle arrest. p14(ARF) possesses multiple binding domains for E2F-1.
Literature	GENERIF_SUMMARY	p73beta is transcriptionally induced by E2F-1 and functions as a positive regulator of apoptosis.
Literature	GENERIF_SUMMARY	Most of the TGF-beta1-induced preapoptotic cells were arrested in G(1) phase of the cell cycle. This was associated with a significant increase in both E2F-DNA-binding activity and transcription of E2F-responsive reporter constructs.
Literature	GENERIF_SUMMARY	E2F-1 regulates the expression of APAF-1 in human cells
Literature	GENERIF_SUMMARY	E2F-1 has a role in regulation of caspase proenzymes through a direct transcriptional mechanism
Literature	GENERIF_SUMMARY	Che-1 contacts the pocket region of Rb and removes HDAC1 from the Rb/E2F1 complex, affecting the activation of E2F-dependent promoters and cell division.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	During thyroid hormone-induced differentiation of embryonic carcinoma cells and of oligodendrocyte precursor cells, levels of E2F-1 mRNA and E2F-1 protein decrease, caused by thyroid hormone receptor regulating the transcription of the E2F-1 gene.
Literature	GENERIF_SUMMARY	IFN-gamma is an effective inhibitor of ASM cell proliferation by blocking transition from G1-to-S phase by acting at two different levels
Literature	GENERIF_SUMMARY	expression of the transcription factor DP-1 and its heterodimeric partner E2F-1 in non-Hodgkin lymphoma
Literature	GENERIF_SUMMARY	We found that E2F1 was present at most of the CpG islands bound by pRb, independent of the phase of the cell cycle, our data suggest that the majority of DNA-bound pRb is recruited to E2F target promoters during both G(0)/G(1) and S phases.
Literature	GENERIF_SUMMARY	P202 inhibits E2F1-mediated apoptosis in prostate cancer cells
Literature	GENERIF_SUMMARY	E2F1 is involved in heat shock-induced cell cycle arrests at the G1/S and G2/M checkpoints, which also may be relevant for hyperthermic cancer therapy
Literature	GENERIF_SUMMARY	esf-1 induces apoptosis and inhibits proliferation of human colon cancer cell lines; a caspase-independent pathway is suggested
Literature	GENERIF_SUMMARY	Chk2 phosphorylates and activates E2F-1 in response to DNA damage, resulting in apoptosis. This suggests a role for E2F-1 in checkpoint control and tumour suppression.
Literature	GENERIF_SUMMARY	In response to DNA damage, E2F1 is directed from cell cycle progression to apoptotic E2F target genes.
Literature	GENERIF_SUMMARY	Analysis of gene expression level profiles showed that parental cell line undergoes apoptosis through an E2F1/p73-dependent pathway while its drug resistant variant evades it.
Literature	GENERIF_SUMMARY	Both senescence-associated heterochromatin foci formation and the silencing of E2F target genes depend on the integrity of the Rb pathway and do not occur in reversibly arrested cells
Literature	GENERIF_SUMMARY	C/EBPepsilon interacts with Rb and E2F1 during granulocytic differentiation
Literature	GENERIF_SUMMARY	E2F1 is upregulated by TR3 orphan nuclear receptor in human prostate cancer cells
Literature	GENERIF_SUMMARY	the Cyclin D1/E2F1 pathway may be involved in apoptotic death of bone marrow cells in myelodysplastic syndromes
Literature	GENERIF_SUMMARY	E2F1 activity is repressed by BRG1, but this repression is blocked when BRG1 interacts with EVI1
Literature	GENERIF_SUMMARY	two E2F-binding sites play distinct roles in the

Main Category	Category	Terms
		regulation of E2F1 transcription by interacting with different sets of E2F members and cooperating with the contiguous repressor element
Literature	GENERIF_SUMMARY	Here we show that the E2F1 binding domain of prohibitin has the potential to fold into a coiled-coil structure.
Literature	GENERIF_SUMMARY	E2F and Sp1/Sp3 synergize but are not sufficient to activate the MYCN gene in neuroblastoma
Literature	GENERIF_SUMMARY	results establish that tumor necrosis factor alpha targets insulin-like growth factor-I induced E2F-transcription facor 1 synthesis, leading to inhibition of accumulation in cyclin A and hyperphosphorylation of RB protein
Literature	GENERIF_SUMMARY	E2F1 transactivation is repressed by MdmX
Literature	GENERIF_SUMMARY	Our results suggest that overexpression of E2F1, induced both by LOH at the RB locus and anomalous phosphorylation of the RB protein, is involved in the development of non-small cell lung carcinoma.
Literature	GENERIF_SUMMARY	Results suggest that E2F1 plays a central role in signaling disturbances in the retinoblastoma growth control pathway and, by upregulation of Chk2 by Atm and Nbs1, may sensitize cells to undergo apoptosis.
Literature	GENERIF_SUMMARY	The data suggest that E2F-1 overexpression plays a role in suppression of tumor, at least in part trough transcriptional regulation of FHIT and relevant activation of WWOX.
Literature	GENERIF_SUMMARY	ErbB3-binding protein Ebp1 binding to E2F promoter elements and E2F-mediated transcription are regulated by heregulin
Literature	GENERIF_SUMMARY	increased expression of the E2F1 gene might play a significant role in human thyroid carcinogenesis through derangement of the Rb-E2F signaling pathway
Literature	GENERIF_SUMMARY	E2F1-dependent recruitment of Tip60 to chromatin occurred in late G(1)
Literature	GENERIF_SUMMARY	E2F1 stabilization in response to DNA damage requires p300/CREB-binding protein-associated factor
Literature	GENERIF_SUMMARY	new roles for several DNA damage response factors by demonstrating that they also participate in the oncogenic stress signaling pathway between E2F1 and p53
Literature	GENERIF_SUMMARY	results reveal an essential role of ACTR in control of breast cancer cell proliferation and implicate the ACTR-E2F1 pathway as a novel mechanism in antiestrogen resistance
Literature	GENERIF_SUMMARY	IL-1beta reduces the ability of IGF-I to activate Cdk2 and to induce E2F-1, cyclin A, and cyclin A-dependent phosphorylation of a retinoblastoma tumor suppressor

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	E2F-1 may play a role in the detection and subsequent repair of damaged DNA [review]
Literature	GENERIF_SUMMARY	mitotic checkpoint protein Mad2 is a direct E2F target and, as a consequence, is aberrantly expressed in cells with Rb pathway defects
Literature	GENERIF_SUMMARY	Thus, expression of dbpA was positively regulated by E2F1, suggesting that one of the effects of E2F1 on cell proliferation might be mediated by dbpA at the carcinogenesis step.
Literature	GENERIF_SUMMARY	Expression of ICBP90 was upregulated by E2F-1, and the upregulation was caused by binding of E2F-1 to the intron1 of ICBP90.
Literature	GENERIF_SUMMARY	repression of hTERT by endogenous p53 is mediated by p21 and E2F
Literature	GENERIF_SUMMARY	RB/E2F signaling pathway activation is required for the modulation of hepatitis C virus core protein-induced cell growth in liver and non-liver cells
Literature	GENERIF_SUMMARY	upregulation of E2F-1 expression by heregulin beta 1 was directly inhibited by HES-1 through the same CACGAG-site as seen with estrogen-stimulated induction
Literature	GENERIF_SUMMARY	Increased expression in myelodysplastic bone marrow cells.
Literature	GENERIF_SUMMARY	the central acidic domain of MDM2 is critical in inhibition of retinoblastoma-mediated suppression of E2F and cell growth
Literature	GENERIF_SUMMARY	The N-terminal 72 amino acids of Pura were involved in E2F-1 binding, inhibition of promoter activation by E2F-1 and reversal of E2F-mediated growth inhibition.
Literature	GENERIF_SUMMARY	E2F-1 expression is inversely correlated with tumor growth in colon adenocarcinoma.
Literature	GENERIF_SUMMARY	there is a negative feedback loop in apoptosis involving E2F and AKT
Literature	GENERIF_SUMMARY	BMP-2 inhibits DHT-induced growth of LNCaP cells through a decrease in E2F protein expression and suppression of E2F activity by hypophosphorylation of Rb
Literature	GENERIF_SUMMARY	induction of p27 acts as a negative feedback mechanism for E2F1 and may also contribute to other functions of E2F1
Literature	GENERIF_SUMMARY	Induction of the metallothionein 1G promoter by VEGF and heavy metals occurs through the utilization of different transcription factors, notably E2F1.
Literature	GENERIF_SUMMARY	We report a novel, noncompetitive mechanism linking acetylation and ubiquitination, in which the association of transcription factor E2F-1 with the cellular coactivator and acetyltransferase p300 determines its

Main Category	Category	Terms
		acetylation and subsequent ubiquitination.
Literature	GENERIF_SUMMARY	expression of E2F1 is negatively regulated by two miRNAs in the cluster, miR-17-5p and miR-20a
Literature	GENERIF_SUMMARY	E2F1/p53 interaction regulates the transcriptional activity of p53 throughout the cell cycle.
Literature	GENERIF_SUMMARY	data identify E2F1 as a critical determinant of the cellular response to death-receptor-mediated apoptosis, and suggest that its downregulation contributes to the immune escape of lung adenocarcinoma tumor cells
Literature	GENERIF_SUMMARY	the expression of the E2F6 repressor is influenced at the transcriptional level by E2F family members and suggest that interplay among these transcriptional regulators, especially E2F1, may be critical for cell cycle regulation
Literature	GENERIF_SUMMARY	A subset of E2F1 targets are instrumental in altering proliferative, differentiation, and apoptotic properties. Deregulation of the E2F/RB pathway may promote the development of hormone-refractory prostate tumors.
Literature	GENERIF_SUMMARY	MDM2 stabilizies E2F1 protein through the E2F1 ubiquitination pathway.
Literature	GENERIF_SUMMARY	E2F1 contributes to the genetic instability associated with transformation and tumorigenesis
Literature	GENERIF_SUMMARY	data show that E2F1 has potential binding activity to the retinoblastoma control element and a different transcriptional regulation pathway which cooperates with the retinoblastoma tumor suppressor protein
Literature	GENERIF_SUMMARY	regulation of p14(ARF) gene by E2F is distinct from that of classical E2F targets. It is directly mediated by E2F through a novel E2F-responsive element that varies from the typical E2F site.
Literature	GENERIF_SUMMARY	E2F1 may participate in telomerase activity regulation in malignant glioma cells. Its expression appears to be strongly associated with the survival of patients with malignant brain tumors.
Literature	GENERIF_SUMMARY	E2F1, in addition to EGFR, target sites induce B-Myb promoter
Literature	GENERIF_SUMMARY	MDM2 has critical roles in the regulation of p21 and E2F1 expression, stability and function [review]
Literature	GENERIF_SUMMARY	Regulation of E2F1 by IKK alpha influences estrogen-mediated cell cycle progression
Literature	GENERIF_SUMMARY	E2F1-induced Mre11-Rad50-Nbs1 (MRN) foci generate a cell cycle checkpoint that with sustained E2F1 activity eventually yields to apoptosis.
Literature	GENERIF_SUMMARY	concomitant induction of E2F1 targets ASK1 and Bim by HDACIs warrants an effective activation of E2F1-dependent apoptosis in response to suberoylanilide hydroxamic acid

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	the release of E2F activity by elevated E2F1 gene copy numbers may play a functional role in melanoma growth
Literature	GENERIF_SUMMARY	Can bind the rRNA promoter and modulate its activity through the interaction with two identifiable E2F1-binding sequences.
Literature	GENERIF_SUMMARY	Up-regulation of ASK-1 may also favour the p53-independent E2F1 apoptotic activity.
Literature	GENERIF_SUMMARY	JPO1/CDCA7 is a unique transcription regulator whose expression is activated by E2F1 as well as c-Myc.
Literature	GENERIF_SUMMARY	Myc and E2F1 engage the ATM signaling pathway to activate p53 and induce apoptosis [review]
Literature	GENERIF_SUMMARY	Results provide insights into the possibility of using E2F1 as a therapeutic target in the treatment of cancer.
Literature	GENERIF_SUMMARY	Deregulation of E2F1 underlies much of the chromosomal instability in breast tumors.
Literature	GENERIF_SUMMARY	only 75 amino acids from within the DNA-binding domain of E2F-1 are sufficient for cell death and that this activity is also present in the DNA-binding domains of E2F-2 and E2F-3
Literature	GENERIF_SUMMARY	histone deacetylase inhibitor sodium butyrate induces G1/S phase arrest in E1A + Ras-transformed cells through down-regulation of E2F1 activity and stabilization of beta-catenin
Literature	GENERIF_SUMMARY	There were no significant changes of CDK4 and E2F-1/4 expression in benzo(a)pyrene treated embryo lung fibroblasts.
Literature	GENERIF_SUMMARY	E2F1 may cooperate with alterations in the MAPK or Rb pathways in the promotion of proliferation in many or most types of melanoma and may become a target for therapeutic intervention.
Literature	GENERIF_SUMMARY	There is a mutual regulation between E2F1 and SirT1 that affects cellular sensitivity to DNA damage.
Literature	GENERIF_SUMMARY	E2F and Wip1 are modulators of E2F1-induced apoptosis involving p38 MAP kinase
Literature	GENERIF_SUMMARY	There is a positive feedback regulatory loop consisting of E2F1 and SRC-3 to maintain high levels of SRC-3 and E2F1 activity, which may partially interpret the oncogenic role of SRC-3 overexpression.
Literature	GENERIF_SUMMARY	Depletion of E2F1 prevented prohibitin from repressing the YY1 promoter.
Literature	GENERIF_SUMMARY	Data demonstrate a novel physiological aspect of E2F1 in human cancer cells, where activated mitochondrial biogenesis occurs as a consequence of the acute loss of E2F1.
Literature	GENERIF_SUMMARY	CDCA4 participates in the regulation of cell

Main Category	Category	Terms
		proliferation, mainly through the E2F/retinoblastoma protein pathway
Literature	GENERIF_SUMMARY	Levels of E2F1 protein were greater in the nucleus of neurons in brains of HIV-infected patients exhibiting dementia when compared to HIV-negative subjects or HIV-positive neurologically normal patients
Literature	GENERIF_SUMMARY	p53 family proteins are potent therapeutic agents for human papilomavirus-associated uterine cervical cancers carrying the E2F1 promoter.
Literature	GENERIF_SUMMARY	offer a new mechanism of regulation of E2F1 by direct binding of glycogen synthase kinase 3 beta to its transactivation domain
Literature	GENERIF_SUMMARY	TFDP3, a novel DP protein, inhibits DNA binding and transactivation by E2F
Literature	GENERIF_SUMMARY	expression of mammalian GINS is regulated by 17beta-Estradiol-stimulated estrogen receptor alpha, and PSF3 acts as a gene responsive to transcription factor E2F1
Literature	GENERIF_SUMMARY	E2F1, E2F2, and E2F3 directly bind the promoter of the mir-17-92 cluster, activating its transcription. miR-20a from the mir-17-92 cluster in turn modulates the translation of E2F2 and E2F3, suggesting an autoregulatory feedback loop.
Literature	GENERIF_SUMMARY	Elevated E2F1, through its ability to repress androgen receptor transcription, may contribute to the progression of hormone-independent prostate cancer.
Literature	GENERIF_SUMMARY	the expression of human ASF1A and ASF1B are upregulated followed by cell proliferation signal, but that of ASF1B is uniquely regulated by transcription factors E2F during cell cycle progression
Literature	GENERIF_SUMMARY	Proliferation-promoting E2F transcription factor E2F-1 plays a pivotal role in the tumor biology of ovarian cancer.
Literature	GENERIF_SUMMARY	The pRb/E2F cell-cycle pathway mediates cell death in Parkinson's disease.
Literature	GENERIF_SUMMARY	E2F-1 is a transcriptional activator of MKP-2 and MKP-2 is an essential cell death mediator in the E2F-1 pathway
Literature	GENERIF_SUMMARY	Overexpression of E2F-1 is associated with neuroendocrine lung tumors
Literature	GENERIF_SUMMARY	Our study demonstrates that E2F-1 mediates apoptosis through transcriptional regulation of PAC1 and subsequent suppression of the ERK signaling
Literature	GENERIF_SUMMARY	Assessment of E2F1 at the mRNA level in primary breast cancer is a strong determinant of breast cancer patient outcome.
Literature	GENERIF_SUMMARY	Critical for hormone regulation of the proliferative program of breast cancer cells.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	LA's ability to inhibit apoptosis and proliferation of ECs could beneficially affect endothelial dysfunction, which precedes manifestation of late diabetic vascular complications.
Literature	GENERIF_SUMMARY	Alien inhibits E2F1 gene expression and cell proliferation.
Literature	GENERIF_SUMMARY	Epstein-Barr virus-encoded latent membrane protein 1inhibited p16(INK4A) expression, promoted phosphorylation of p105 Rb and upregulated E2F1 expression and overexpression of E2F1 alone was sufficient to upregulate telomerase activity.
Literature	GENERIF_SUMMARY	Results show that p14ARF regulates E2F-1 ubiquitination and degradation via a p53-dependent mechanism.
Literature	GENERIF_SUMMARY	Chk1 is E2F regulated and highly expressed in triple-negative estrogen receptor /progesterone receptor / HER-2 breast carcinomas
Literature	GENERIF_SUMMARY	E2F-1 is a transcriptional activator of DUSP1 and DUSP1 is a link between E2F-1 and MAP kinases
Literature	GENERIF_SUMMARY	The mutation and expression of E2F-1 in human gastric cancer tissues and the effect of E2F-1 overexpression on the proliferation of gastric carcinoma cells, was investigated.
Literature	GENERIF_SUMMARY	ectopically BRCA2-expressing cells have different intracellular levels of Aurora A, Aurora B, p21, E2F-1, and pRb, suggesting a BRCA2-mediated suppression of polyploidy via stabilization of the checkpoint proteins levels
Literature	GENERIF_SUMMARY	by disrupting the DNA binding activity of E2F1, BTG3 participates in the regulation of E2F1 target gene expression. Therefore, our studies have revealed a previously unidentified pathway through which the activity of E2F1 may be guarded by activated p53.
Literature	GENERIF_SUMMARY	E2F-1 protein overexpression can be useful as a prognostic indicator in epithelial ovarian neoplasms.
Literature	GENERIF_SUMMARY	KAP1 contributes to the negative regulation of E2F1 and may serve as a partial backup to prevent E2F1-mediated apoptosis in the absence of pRb
Literature	GENERIF_SUMMARY	NF-kappaB recruits E2F1 to fully activate the transcription of NF-kappaB target genes.
Literature	GENERIF_SUMMARY	A hitherto unrecognized mechanism to control E2F1 function through modulation of its subcellular localization, is identified.
Literature	GENERIF_SUMMARY	the E2F1 transcription factor, which is implicated in carcinoma invasiveness, upregulates the expression of PACE4
Literature	GENERIF_SUMMARY	Transcriptional regulation of a new variant of PDGFRA by E2F1.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	E2F1binding sites are located within 2 kb of a transcription start site, in both normal and tumor cells
Literature	GENERIF_SUMMARY	Clear cell renal cell carcinoma is characterized by the overexpression of E2F1, which is likely a result of a deregulated control of T3-dependent molecular processes.
Literature	GENERIF_SUMMARY	Human herpesvirus 6A (HHV-6A) and HHV-6B infection was associated with significant reduction of E2F1/Rb complexing, altered E2F1 localization and cell cycle arrest in T cells.
Literature	GENERIF_SUMMARY	Expression of the transcription factor E2F1 is upregulated by addition of Clonorchis sinensis excretory-secretory products in HEK293 cells.
Literature	GENERIF_SUMMARY	RhoBTB2 up-regulated during drug-induced apoptosis, with this being primarily dependent on E2F1.
Literature	GENERIF_SUMMARY	E2F1 gene expression correlates with thymidylate synthase and survivin gene expressions and tumor proliferation. During the progression of NSCLC, E2F1 overexpression could produce more aggressive tumors with a high proliferation rate and chemoresistance.
Literature	GENERIF_SUMMARY	A subset of E2F1 target genes that are specifically repressed by PI3K/Akt signaling, thus distinguishing the E2F1 proliferative or apoptotic function, is identified.
Literature	GENERIF_SUMMARY	E2F-1 is a potentially novel independent prognostic factor that may identify gastric cancer patients who will likely benefit from adjuvant chemoradiation therapy following curative resection.
Literature	GENERIF_SUMMARY	CARM1 is a critical factor in the pathway of estrogen-stimulated breast cancer growth downstream of ERA and AIB1and upstream of the cell cycle regulatory transcription factor E2F1.
Literature	GENERIF_SUMMARY	E2F1 stimulates ataxia telangiectasia mutated (ATM) kinase-dependent p53 phosphorylation and apoptosis through a mechanism that does not involve DNA damage.
Literature	GENERIF_SUMMARY	These studies identify E2F1 as a positive transcriptional regulator for delta-catenin.
Literature	GENERIF_SUMMARY	Suggest that the miR-106b-25 cluster is involved in E2F1 posttranscriptional regulation and may play a key role in the development of TGFbeta resistance in gastric cancer.
Literature	GENERIF_SUMMARY	Myc is required to allow the interaction of the E2F1 protein with the E2F gene promoters.
Literature	GENERIF_SUMMARY	Brip1 is a genuine target gene for the E2F/Rb pathway; elevated expression levels of Brip1 are detected in primary invasive breast carcinomas with unfavorable characteristics.
Literature	GENERIF_SUMMARY	Results suggest that Bin1 gene suppression caused by

Main Category	Category	Terms
		oncogenic E1A via Rb/E2F1 inactivation is an essential step in cell cycle progression promoted by c-Myc, and subsequently, E1A transformation.
Literature	GENERIF_SUMMARY	E2F1 contributes to the transcriptional activation of the KIR3DL1 gene.
Literature	GENERIF_SUMMARY	The regulatory effect of E2F1 suggests that DYRK1A may play a role in cell cycle regulation or apoptosis.
Literature	GENERIF_SUMMARY	These results suggest that E2F1 plays an important role in regulating RANKL transcription through binding to the E2F consensus binding site.
Literature	GENERIF_SUMMARY	EBNA-6 binds to a MRPS18-2 protein, and targets it to the nucleus, where MRPS18-2 competes with the binding of E2F1 to pRb, thereby raising the level of free E2F1, and lifting the block preventing S-phase entry.
Literature	GENERIF_SUMMARY	AP-1 activity and cyclin D1/E2F-1 expression in human embryo lung fibroblasts increased significantly after benzo(a)pyrene treatment.
Literature	GENERIF_SUMMARY	The current findings demonstrated that loss of Rb and p16/INKa expression and high E2F1 expression indicate impairment of the Rb suppressor pathway.
Literature	GENERIF_SUMMARY	ASK/Dbf4, a novel cell survival gene in melanoma is transcriptionally regulated by E2F1.
Literature	GENERIF_SUMMARY	DAPK2 as a novel Sp1-dependent target gene for E2F1 and KLF6 in cell death response.
Literature	GENERIF_SUMMARY	AHR binds to E2F1 and inhibits E2F1-induced apoptosis
Literature	GENERIF_SUMMARY	E2F6 has a role in control of hypoxia-induced apoptosis through regulation of E2F1
Literature	GENERIF_SUMMARY	PTPRO gene is co-regulated by both E2F1 and miR-17-92.
Literature	GENERIF_SUMMARY	E2F may regulate G2/M transition through the induction of SIL. Furthermore, as silencing of SIL cause apoptosis in cancer cells, these finding may have therapeutic relevance in tumors with constitutive activation of E2F
Literature	GENERIF_SUMMARY	E2F1, not Par-4, was found to be directly bound to the Smac promoter, suggesting that Par-4 exerted indirectly its transcriptional control on the Smac gene though interacting with E2F1.
Literature	GENERIF_SUMMARY	Data show that MCPH1 cooperates with E2F1 to regulate genes involved in DNA repair, checkpoint and apoptosis, and might participate in the maintenance of genomic integrity.
Literature	GENERIF_SUMMARY	activation of E1AF provides a means for E2F1 to induce cell apoptosis in response to DNA damage
Literature	GENERIF_SUMMARY	E2F-1 and COX-2 are overexpressed in oral cancer
Literature	GENERIF_SUMMARY	E2F1 may function as a critical antiapoptotic factor

Main Category	Category	Terms
		both in human and in rodent liver cancer through its ability to counteract c-Myc-driven apoptosis via activation of PIK3CA/Akt/mTOR and c-Myb/COX-2 pathways
Literature	GENERIF_SUMMARY	The levels of E2F1 mRNA were lower in malignant breast tissues. They declined further with increasing TNM stage. Results suggest E2F1 is a tumour suppressive gene in human breast cancer.
Literature	GENERIF_SUMMARY	E2F1 is a potent and specific inhibitor of beta-catenin/T-cell factor (TCF)-dependent transcription, and that this function contributes to E2F1-induced apoptosis
Literature	GENERIF_SUMMARY	E2F1 controls pre-mRNA processing events to induce apoptosis and identify the SC35 Ser-Rich Arg protein as a key direct E2F1-target in this setting.
Literature	GENERIF_SUMMARY	paclitaxel induces apoptosis in human retinoblastoma Y79 cells by up-regulating E2F1
Literature	GENERIF_SUMMARY	These data provide a mechanistic view of miRNA-based regulation of E2F1 in the context of the emerging model that miRNAs coordinate the timing of cell cycle progression.
Literature	GENERIF_SUMMARY	analysis of transcriptional repression of the prosurvival endoplasmic reticulum chaperone GRP78/BIP by E2F1
Literature	GENERIF_SUMMARY	study reports that mitotic complex genes Ect2, RacGAP, and MKLP1 are coordinately induced in S phase in proliferating T lymphocytes as well as in epithelial cells, depending upon activity of the CUX1 and E2F1 transcription factors
Literature	GENERIF_SUMMARY	deregulation of c-MYC in lymphoma cells does not overcome the tumor suppressor function of TGF-beta and that repression of E2F-1 transcription is sufficient for the efficient induction of cytostasis.
Literature	GENERIF_SUMMARY	E2F1 could upregulate endogenous XRCC1 expression and stimulate the XRCC1 promoter.
Literature	GENERIF_SUMMARY	expression of miRNAs is downregulated in senescent cells and in breast cancers harboring wild-type p53. These miRNAs are repressed by p53 in an E2F1-mediated manner
Literature	GENERIF_SUMMARY	Bortezomib arrests the proliferation of hepatocellular carcinoma cells HepG2 and JHH6 by differentially affecting E2F1 level.
Literature	GENERIF_SUMMARY	These results suggest that the p27(Kip1) gene specifically senses deregulated E2F activity through the E2F-like elements to suppress inappropriate cell cycle progression in response to loss of pRb function.
Literature	GENERIF_SUMMARY	under basal nonapoptotic conditions that NF-kappaB constitutively occupies and transcriptionally silences Bnip3 gene transcription by competing with E2F-1
Literature	GENERIF_SUMMARY	overlapping E2F1/Sp1 site, being present in multiple

Main Category	Category	Terms
		copies in the p14ARF promoter, may serve as the targets for both E2F1 and Sp1, thereby playing a crucial role in response to some oncogenic signals and stimulators
Literature	GENERIF_SUMMARY	E2F1 participates in the regulation of the transcriptional activation of the KIR3DL1 gene.
Literature	GENERIF_SUMMARY	The release of PCAF from hSirT1 repression favors the assembly of transcriptionally active PCAF/E2F1 complexes onto the P1p73 promoter and p53-independent apoptosis.
Literature	GENERIF_SUMMARY	E2F1 activated the expression of transcription factor EGR1 for promoting cell survival.
Literature	GENERIF_SUMMARY	Overexpression of E2F-1 significantly inhibits cell growth and proliferation, blocking entry into the S-phase of the cell cycle.
Literature	GENERIF_SUMMARY	Transcriptional regulation of human fibroblast growth factor receptor 1 by E2F1 is reported.
Literature	GENERIF_SUMMARY	Nutlin-3 treatment upregulated TAp73 and E2F1 protein levels.
Literature	GENERIF_SUMMARY	E2F1 transcription is induced by genotoxic stress through ATM/ATR activation.
Literature	GENERIF_SUMMARY	E2F1 is highly expressed in Burkitt's lymphoma cell lines and sBL lymphoma specimens. Reduction of its expression in sBL cells inhibits tumor formation and decreases their proliferation rate.
Literature	GENERIF_SUMMARY	Letter
Literature	GENERIF_SUMMARY	Cdh1 reciprocally regulates the Rb pathway through competing with E2F1 to bind the hypophosphorylated form of Rb.
Literature	GENERIF_SUMMARY	Transcription factors were screened for regulation of a human HIC1 promoter reporter. Found that E2F1 strongly activates the full-length HIC1 promoter reporter.
Literature	GENERIF_SUMMARY	acetylation status of E2F-1 contributes to the selective activation of tumor suppressor p73 gene.
Literature	GENERIF_SUMMARY	high E2F1 levels exerted a growth-suppressing effect in osteosarcomas that relied on the integrity of the DNA damage response network
Literature	GENERIF_SUMMARY	present study indicates that E2F1 has a dual role in the regulation of p73.
Literature	GENERIF_SUMMARY	miR-330 induces apoptosis in prostate cancer cells through E2F1-mediated suppression of Akt phosphorylation.
Literature	GENERIF_SUMMARY	Investigated downregulation of coronary vascular smooth muscle cell growth by siRNA against E2F1, cyclins E1 and E2. Data shows reduction in the

Main Category	Category	Terms
		phosphorylation levels of the retinoblastoma protein pRB and a decrease in the amount of cyclin A2.
Literature	GENERIF_SUMMARY	BIN1 is transcriptionally activated by E2F1 and mediates E2F1-induced apoptosis in response to DNA damage.
Literature	GENERIF_SUMMARY	E2F1 expression induced a significant increment in endogenous IGF-IR levels. ChIP assays showed enhanced E2F1-binding to the IGF-IR promoter in E2F1-expressing cells.
Literature	GENERIF_SUMMARY	CK1alpha plays a central role in mediating MDM2 control of p53 and E2F-1 protein stability
Literature	GENERIF_SUMMARY	Sequence changes in the E2F1 HCF-1-binding site can modulate both up and down the ability of E2F1 to induce apoptosis indicating that HCF-1 association with E2F1 is a regulator of E2F1-induced apoptosis.
Literature	GENERIF_SUMMARY	analysis of the transcriptional axis mutant p53, E2F1 and ID4 which promotes tumor neo-angiogenesis
Literature	GENERIF_SUMMARY	E2F1 transcription factor negatively regulates human IRF-3 gene promoter.
Literature	GENERIF_SUMMARY	reveals a tumor suppressor function of miR-449a/b through regulating Rb/E2F1 activity, and suggests that escape from this regulation through an aberrant epigenetic event contributes to E2F1 deregulation and unrestricted proliferation in human cancer
Literature	GENERIF_SUMMARY	Results elucidate a mechanism for abundant tumor hPTTG1 expression, whereby Rb inactivation releases E2F1 to induce hPTTG1. This signaling pathway may underlie the requirement of PTTG1 for pituitary tumorigenesis.
Literature	GENERIF_SUMMARY	Low E2F1 is associated with lymph node metastases in breast carcinoma.
Literature	GENERIF_SUMMARY	Data show that E2F1 induces EZH2 expression, which in turn antagonizes the induction of E2F1 pro-apoptotic target Bim expression.
Literature	GENERIF_SUMMARY	Overexpression of E2F-1 accelerates apoptosis of gastric carcinoma MGC-803 cells.
Literature	GENERIF_SUMMARY	Data suggest that miR-449a and b provide a twofold safety mechanism to avoid excessive E2F1-induced proliferation by cell cycle arrest and by apoptosis.
Literature	GENERIF_SUMMARY	E2F1-epidermal growth factor receptor interaction is a driving force in melanoma progression that may serve as a paradigm for E2F1-induced metastasis in other human cancers.
Literature	GENERIF_SUMMARY	miR-223 is down-regulated in different subtypes of acute myeloid leukemia (AML). E2F1 binds to the miR-223 promoter in AML blast cells and inhibits miR-223 transcription.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	There are specific interactions among Rb, HDAC1 and E2F1 proteins in gallbladder carcinoma, indicating the existence of the direct Rb/E2F1/HDAC1 signal transduction pathway. p16 does not interact with any of these proteins.
Literature	GENERIF_SUMMARY	There is over-expression of E2F-1 gene and deletion of p14(ARF) gene in the tumorigenesis of the lung cancer in Gejiu and Xuanwei regions in Yunnan Province.
Literature	GENERIF_SUMMARY	E2F1 makes use of its transcriptional target RRP1B to activate other genes directly involved in apoptosis.
Literature	GENERIF_SUMMARY	CDKN1C negatively regulates RNA polymerase II C-terminal domain phosphorylation in an E2F1-dependent manner
Literature	GENERIF_SUMMARY	Data show that progesterone receptor-B acts in both direct and indirect manners to positively upregulate E2F1 expression.
Literature	GENERIF_SUMMARY	that E2F1 can activate transcription from the Siah1 promoter. Ectopic expression of E2F1 elevates the Siah1 level, hence suppressing the beta-catenin/TCF activity
Literature	GENERIF_SUMMARY	Collectively, our findings suggest that tamoxifen resistance is a result of increased ERalpha/Sp-1 interaction, which enhances the expression of E2F1 to promote tamoxifen resistance.
Literature	GENERIF_SUMMARY	Microarray expression analysis indicated that E2F1 and E2F2 are downstream genes in the KDM5B pathway.
Literature	GENERIF_SUMMARY	Cis-acting genetic variation at an E2F1/YY1 response site and putative p53 site is associated with altered allele-specific expression of ERCC5 (XPG) transcript in normal human bronchial epithelium.
Literature	GENERIF_SUMMARY	In the absence of Rb, E2F1 overexpression results in apoptosis, not proliferation, and that this effect is independent of p53.
Literature	GENERIF_SUMMARY	AR mediates EGF- and bFGF-induced colon cancer cell proliferation by activating or expressing G(1)-S phase proteins such as E2F-1, cdks, and cyclins through the reactive oxygen species/phosphoinositide 3-kinase/AKT pathway.
Literature	GENERIF_SUMMARY	This study shows that RIP140 is a regulator of the E2F pathway, which discriminates luminal- and basal-like tumors
Literature	GENERIF_SUMMARY	Lactoferrin exploits the control mechanism of E2F1-regulated target genes or Bcl-2 family gene networks involved in the apoptotic process in Jurkat human leukemia T lymphocytes.
Literature	GENERIF_SUMMARY	E2F1 has a direct, non-transcriptional role in DNA repair involving increased recruitment of NER factors to sites of damage

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Supervised analysis of gene expression data revealed a gene expression signature that is strongly associated with invasive bladder tumors.
Literature	GENERIF_SUMMARY	E2F1 is a critical regulator of dendritic cell maturation.
Literature	GENERIF_SUMMARY	Transcriptional analysis of an E2F gene signature as a biomarker of activity of the cyclin-dependent kinase inhibitor PHA-793887 in tumor and skin biopsies.
Literature	GENERIF_SUMMARY	These results suggest that phosphorylation of Ser403 could influence the selection and regulation of E2F1 target genes.
Literature	GENERIF_SUMMARY	these data highlight an important role of miR-331-3p in cell cycle control by targeting 3'-UTR of cell cycle-related molecule E2F1 in gastric cancer.
Literature	GENERIF_SUMMARY	beta1,4GalT I is an important E2F1 target gene that is required for cell cycle progression in mammalian cells, which elicits a new mechanism of cell growth and a new mechanism of beta1,4GalT I transcription
Literature	GENERIF_SUMMARY	Downregulation of the repair genes is on account of a decrease in amount and promoter recruitment of the E2F1 transcription factor
Literature	GENERIF_SUMMARY	Set9 and LSD1 regulate DNA damage-induced cell death in a manner opposite to that observed in p53(+/+) cells, via modulation of E2F1 stabilization.
Literature	GENERIF_SUMMARY	Our findings not only provided the first evidence that Akt1 is a direct target of miRNA but also demonstrated that miR-149* is a pro-apoptotic miRNA by repressing the expression of Akt1 and E2F1.
Literature	GENERIF_SUMMARY	Results indicate that co-regulation of ANRIL and p14ARF could be coupled by their unique intergenic region potentially through E2F1.
Literature	GENERIF_SUMMARY	E2F-1 associated pRB is modified by O-GlcNAc.
Literature	GENERIF_SUMMARY	Coordination between cell cycle progression and cell fate decision by the p53 and E2F1 pathways in response to DNA damage.
Literature	GENERIF_SUMMARY	RB.E2F1 complex plays a critical role in ZBRK1 transcriptional repression, and loss of this repression may contribute to cellular sensitivity of DNA damage, ultimately leading to carcinogenesis.
Literature	GENERIF_SUMMARY	TRIM16 acts as a tumour suppressor, affecting neuritic differentiation, cell migration and replication through interactions with cytoplasmic vimentin and nuclear E2F1 in neuroblastoma cells.
Literature	GENERIF_SUMMARY	E2F1 binds to some consensus binding sequences in the MDM2 promoter, but this direct binding is dispensable for the inhibition of MDM2 promoter activity by E2F1.
Literature	GENERIF_SUMMARY	Data report that ANCCA directly interacts with E2F1 to

Main Category	Category	Terms
		E2F3 in its complex with MLL and that its N terminus interacts with both the N and C termini of E2F1.
Literature	GENERIF_SUMMARY	These data suggest that APC/C(Cdc20) specifically targets E2F1 for degradation in early mitosis and reveal a novel mechanism for limiting free E2F1 levels in cells, failure of which may compromise cell survival and/or homeostasis.
Literature	GENERIF_SUMMARY	Findings establish that NF-Y acts upstream of E2F1 in p53-mediated apoptosis.
Literature	GENERIF_SUMMARY	Enhanced NPM-retinoblastoma tumor suppressor protein (pRB) interaction, leading to the relief of the repressive pRB-E2F1 circuitry and the consequent transcriptional activation of E2F1 and several downstream DNA repair genes.
Literature	GENERIF_SUMMARY	CEBPD reverses RB/E2F1-mediated gene repression and participates in HMDB-induced apoptosis of cancer cells.
Literature	GENERIF_SUMMARY	Data show that Orc1 specifically binds hypo-phosphorylated Rb and that this interaction is competitive with the binding of Rb to E2F1.
Literature	GENERIF_SUMMARY	A new function for these factors in the control of insulin secretion and open up new avenues for the treatment of diabetes.
Literature	GENERIF_SUMMARY	The studies support coordinated regulation of delta-catenin expression by both the activating transcription factor E2F1 and repressive transcription factor Hes1 in prostate cancer progression.
Literature	GENERIF_SUMMARY	The transcription factor E2F1 negatively regulates the basal transcriptional activity of IRF-3 and deregulates IRF-3 expression at the mRNA level.
Literature	GENERIF_SUMMARY	E2F1 emerges as a key protein that integrates cell division and growth, both of which are essential for cell proliferation.
Literature	GENERIF_SUMMARY	Mir-17, -20a, and -106b downregulate a common set of pro- and anti-proliferative target genes to impact cell cycle progression of human cord blood stem cells and increase intracellular activity of E2F transcription factors to govern G(1)/S transition.
Literature	GENERIF_SUMMARY	new paradigm for the regulation of cell survival by highlighting the close proximity between the cell cycle factors E2F-1 and NF-kappaB and regulation of the intrinsic death pathway through the expression of Bnip3.
Literature	GENERIF_SUMMARY	Findings do not provide support for the model that protein-protein interactions are involved in recruiting E2F1 to the genome, but rather suggest that recognition of a motif found at most human promoters is the critical determinant.
Literature	GENERIF_SUMMARY	E2F1 is dynamically regulated by epigenetic enzymes in

Main Category	Category	Terms
		response to DNA damage.
Literature	GENERIF_SUMMARY	Owing to the role of B-Myb and E2F1 transcription factors in controlling cell-cycle progression of leukemic cells, the downregulation of these oncogenes by miR-34a suggests the usefulness of therapeutic approaches aimed to modulate the levels of miR-34a.
Literature	GENERIF_SUMMARY	E2F1 plays a crucial role in transcriptional control of the human Lsh gene and the decrease of Lsh expression in senescent cells is related to the repression of E2F1
Literature	GENERIF_SUMMARY	our data identify the miR-15 and miR-16 families as novel transcriptional targets of E2F, which, in turn, modulates E2F activity.
Literature	GENERIF_SUMMARY	miRNA-205 suppresses melanoma cell proliferation and induces senescence via regulation of E2F1 protein
Literature	GENERIF_SUMMARY	Silencing the NHE1 gene significantly inhibited the hypoxia-induced proliferation, hypertrophy, and migration of human pulmonary artery smooth muscle cells via repression of the nuclear transcription factor E2F1.
Literature	GENERIF_SUMMARY	overexpression of ASPP1 rendered MCF-7 and MDA-MB231 breast cancer cells more sensitive to resveratrol-mediated apoptosis via the E2F pathway
Literature	GENERIF_SUMMARY	Studies suggest that BAP1 influences cell proliferation at G1/S by co-regulating transcription from HCF-1/E2F-governed promoters.
Literature	GENERIF_SUMMARY	Transcription factor E2F1 promotes the recruitment of DNA repair factors to sites of DNA double-strand breaks.
Literature	GENERIF_SUMMARY	ICAT is a direct transcriptional target of E2F1, and activation of ICAT by E2F1 is required for E2F1 to inhibit beta-catenin activity.
Literature	GENERIF_SUMMARY	PI3-kinase/p38 kinase-dependent E2F1 activation is critical for Pin1 induction in tamoxifen-resistant breast cancer cells.
Literature	GENERIF_SUMMARY	These results reveal a novel role for E2F1 in mediating an ataxia telangiectasia mutated (ATM) protein kinase-dependent DNA damage responses (DDRs) that contributes to viral replication.
Literature	GENERIF_SUMMARY	results suggest that, in cancer, frequent and persistent expression of PAX8 is required for cell growth control through transcriptional activation of E2F1 expression and upregulation of the RB-E2F1 pathway
Literature	GENERIF_SUMMARY	Findings define a MKI67, TOP2A, and E2F1 tri-marker proliferation index with possible utility for improved prognostication and treatment stratification in prostate cancer.
Literature	GENERIF_SUMMARY	one of the functions of cIAP1 when localized in the nucleus is to regulate E2F1 transcriptional activity.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	miR-106a can increase p53 expression via E2F1 inhibition.
Literature	GENERIF_SUMMARY	Data suggest E3 ligase murine double minute-2 (MDM2) as a drug target in melanoma cells, and identify E2F1 as a biomarker to consider when stratifying putative candidates for clinical studies of p53-MDM2 inhibitors.
Literature	GENERIF_SUMMARY	Integrin subunits alpha5 and alpha6 regulate cell cycle by modulating the chk1 and Rb/E2F pathways to affect breast cancer metastasis.
Literature	GENERIF_SUMMARY	E2F1 directly transcribes QKI, which, in turn, negatively regulates the cell cycle by targeting multiple cell cycle regulators, forming an E2F1-QKI-pRb/E2F1 negative feedback loop.
Literature	GENERIF_SUMMARY	gamma-Tubulin mutants showed that the C terminus interacts with the transcription factor E2F1 and that the E2F1-gamma-tubulin complex is formed during the G(1)/S transition
Literature	GENERIF_SUMMARY	Tamoxifen resistant tumors displayed enriched expression of genes related to cell cycle and proliferation, as well as elevated activity of E2F transcription factors.
Literature	GENERIF_SUMMARY	p16(INK4a) is also a modulator of transcription and apoptosis through controlling the expression of two major transcription regulators, AUF1 and E2F1
Literature	GENERIF_SUMMARY	Interference of POLR1A inhibited the synthesis of rRNA and hindered cell cycle progression in cells with inactivated p53, as a consequence of downregulation of the transcription factor E2F-1.
Literature	GENERIF_SUMMARY	the first somatic mutation of E2F1, R166H, to be identified in human mesothelioma of the peritoneum; R166H mutation in E2F1 is shown to have a deleterious effect on its DNA binding ability
Literature	GENERIF_SUMMARY	Results suggest that DNMT1 operates either as a functional intermediary or in cooperation with E2F1 inhibiting AR gene expression in a methylation independent manner.
Literature	GENERIF_SUMMARY	sequence comparison of vertebrate species suggested that the identified E2F1 binding motif is primate specific, consistent with the previous observation of rapid evolution of MCPH1 protein sequence in primates
Literature	GENERIF_SUMMARY	Tip60 interacts with E2F1 and promotes its acetylation.
Literature	GENERIF_SUMMARY	DNA damage signals alter pRB and E2F1 to engage them in functions leading to apoptotic induction that are distinct from pRB-E2F regulation in cell cycle control
Literature	GENERIF_SUMMARY	Studies indicate that the tumor suppressor E2F1 transcription factor (E2F1) is most relevant for cancer progression and chemoresistance.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	These findings support the hypothesis that Epstein-Barr virus EBNA1 initiates transcription at the C promoter via interactions between multiple EBNA1 homodimers and cellular transcription such as E2F1, ARID3A and Oct-2.
Literature	GENERIF_SUMMARY	E2F1 suppresses TLR3 transcription, but during immune stimulation, Rb is upregulated to block the inhibitory effect of E2F1 on TLR3, highlighting a role of Rb-E2F1 axis in the innate immune response in epithelial cells
Literature	GENERIF_SUMMARY	The results showed that arginine methylation regulates the biological activity of E2F1 activity, and raise the possibility that arginine methylation contributes to tumourigenesis by influencing the E2F pathway.
Literature	GENERIF_SUMMARY	E2F1 and E2F2 genetic variants may jointly play important roles in head and neck squamous cell carcinoma
Literature	GENERIF_SUMMARY	NuSAP is a novel biomarker for prostate cancer recurrence after surgery and its overexpression appears to be driven in part by E2F1 activation
Literature	GENERIF_SUMMARY	we show that the increased expression of E2F-1 protein in human hepatocellular carcinoma is correlated with enhanced tumor cell apoptosis
Literature	GENERIF_SUMMARY	E2F1 is involved in the induction of p19INK4d following UV irradiation
Literature	GENERIF_SUMMARY	E2F-1 participates in p53-mediated DNA damage response and might have a checkpoint function to limit overactive p53.
Literature	GENERIF_SUMMARY	Data show that TFDP3 upregulates the expression of autophagy gene LC3B and inhibits E2F1-induced apoptosis, and may play an important role in prostate cancer.
Literature	GENERIF_SUMMARY	The results suggest that MIC-1 secreted from cancer cells stimulates endothelial cell proliferation by enhancing AP-1- and E2F-dependent expression of G(1) cyclins via PI3K/Akt, JNK, and ERK signaling pathways.
Literature	GENERIF_SUMMARY	results suggest that oncolytic adenovirus induces autophagic cell death through an E2F1-miR-7-EGFR pathway in human cancer cells, providing a novel insight into the molecular mechanism of an anticancer virotherapy
Literature	GENERIF_SUMMARY	Down-regulation of miRNA targeting HMGA1, HMGA2, and E2F1 in human pituitary GH adenomas.
Literature	GENERIF_SUMMARY	The formation of Ub-K11 chains on E2F1 is increased in the presence of Cdh1 and accumulated in the presence of proteasome inhibitor, suggesting that APC/C (Cdh1) targets E2F1 for degradation by forming Ub-K11 chains.
Literature	GENERIF_SUMMARY	this work identifies the RIP140 gene as a new transcriptional target of E2F1 which may explain some of the effect of E2F1 in both cancer and metabolic

Main Category	Category	Terms
		diseases.
Literature	GENERIF_SUMMARY	This report identifies two additional PTPs, SHP-2 and PTP-PEST, that are also directly activated by the Rb-E2F pathway and which can contribute to signal transduction during p53-independent apoptosis.
Literature	GENERIF_SUMMARY	The inverse immunohistochemical relationship between E2F1 and E2F4 indicates a possible mechanistic interlink in colorectal cancer.
Literature	GENERIF_SUMMARY	Direct cooperation between androgen receptor and E2F1 reveals a common regulation mechanism for androgen-responsive genes in prostate cells.(
Literature	GENERIF_SUMMARY	TIGAR mediates de-phosphorylation of RB and stabilisation of RB-E2F1 complex.
Literature	GENERIF_SUMMARY	RBM38 as novel transcriptional target of E2F1 restricts E2F1-induced proliferation. Furthermore, this negative feedback loop seems to restrict tumor aggressiveness, thereby promoting survival of patients with cancer.
Literature	GENERIF_SUMMARY	E2F1 plays a key role in promoting FOXM1 expression, cell survival, and epirubicin resistance as its depletion by siRNA attenuated FOXM1 induction and cell viability in response to epirubicin.
Literature	GENERIF_SUMMARY	The transactivation domain of E2F-1 is not essential for induction of autophagy; apoptosis and autophagy cooperate for an efficient cancer cell killing effect induced by E2F-1 or E2Ftr.
Literature	GENERIF_SUMMARY	E2F1 overexpression diminishes ATM transcription after doxorubicin exposure which is impaired by E2F1 dominant negative mutants.
Literature	GENERIF_SUMMARY	E2F-1 is a substrate for NEDD8 post-translational modification.
Literature	GENERIF_SUMMARY	up-regulation of the E2F1/CDK1 pathway that is associated with neuronal apoptosis following impact spinal cord injury
Literature	GENERIF_SUMMARY	E2F1 and miR-205 are crucial regulators of genes associated with chemoresistance in aggressive tumors.
Literature	GENERIF_SUMMARY	E2F1 up-regulates the human CD2AP promoter.
Literature	GENERIF_SUMMARY	Upregulation of E2F1 is associated with hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	The phosphorylation of S375 by CDK8 regulates E2F1 ability to repress transcription of beta-catenin/TCF-dependent genes, as well as activation of E2F1-dependent genes.
Literature	GENERIF_SUMMARY	p53 inactivation upregulates p73 expression through E2F-1 mediated transcription
Literature	GENERIF_SUMMARY	Novel molecular mechanisms for the different expression patterns of E2F1 targets and the selectivity

Main Category	Category	Terms
		of retinoblastoma protein-mediated heterochromatin formation and gene repression during cellular senescence.
Literature	GENERIF_SUMMARY	Data indicate that overexpression of SENP8, a NEDD8-specific cysteine protease, resulted in deNEDDylation of E2F1 and promoted its transactivation activity at the p73 gene.
Literature	GENERIF_SUMMARY	DNA methylation of the DOK1 core promoter region found in head and neck cancer cell lines hampered the recruitment of E2F1 to the DOK1 promoter and compromised DOK1 expression.
Literature	GENERIF_SUMMARY	High E2F1 expression is associated with B-chronic lymphocytic leukemia.
Literature	GENERIF_SUMMARY	The results identify E2F8 as a repressor and E2F1 as an activator of a transcriptional network controlling polyploidization in mammalian cells.
Literature	GENERIF_SUMMARY	Reduction of miR-26a expression leads to increased Rb1 protein level which in turn inhibits the function of E2F1.
Literature	GENERIF_SUMMARY	Data show an unexpected dual downregulation of both miRNA 17-5p and its target gene E2F1 on the mRNA and protein levels in pediatric systemic lupus erythematosus.
Literature	GENERIF_SUMMARY	Transfection of the NSCLC cell lines with specific siRNAs against SRF, E2F1 and survivin resulted in a very considerable reduction of the intracellular mRNA concentration.
Literature	GENERIF_SUMMARY	Bim, a proapoptotic protein, up-regulated via transcription factor E2F1-dependent mechanism, functions as a prosurvival molecule in cancer
Literature	GENERIF_SUMMARY	Knockdown of E2F1 stabilized HIF-1alpha and promoted LOX expression, while knockdown of both E2F1 and HIF-1alpha prevented the up-regulation of LOX
Literature	GENERIF_SUMMARY	Data gathered from cell lines, tumorigenicity studies, and primary hepatocellular carcinoma samples demonstrate a negative role of HIF-2alpha in tumors, which is mediated by the TFDP3/E2F1 pathway.
Literature	GENERIF_SUMMARY	that the serine-threonine kinase PBK/TOPK is frequently overexpressed in high-grade lymphomas and its expression is positively correlated to that of c-Myc and E2F1
Literature	GENERIF_SUMMARY	Data show that E2F1 overexpression, due to p53 or p21 inactivation, promotes expression of human oncoprotein CIP2A.
Literature	GENERIF_SUMMARY	Knocking down E2F1 reduced the effects of core promoter mutations on SKP2 and cellular proliferation
Literature	GENERIF_SUMMARY	found that HBXIP was able to stimulate the promoter of

Main Category	Category	Terms
		Skp2 through binding to the -640/-443 region in Skp2 promoter involving activating E2F transcription factor 1
Literature	GENERIF_SUMMARY	inhibition of Bcl-2 and Bcl-xL by ABT-737 enhances a caspase-dependent increase in Noxa induction mediated by E2F-1 and, less expectedly, by its regulator protein pRb.
Literature	GENERIF_SUMMARY	Data suggest that genotoxic drug treatment favors selection of chemoresistant cells in genetically heterogeneous tumors, in a manner requiring dysregulation of incoherent feedforward loops that involve E2F1, p73/DNp73, and miR-205.
Literature	GENERIF_SUMMARY	Results suggest that AIB1 promotes BC cell proliferation through the AKT pathway and E2F1. Furthermore, AIB1 overexpression predicts tumour progression in patients with non-muscle-invasive BC.
Literature	GENERIF_SUMMARY	Independently of EAPP, E2F1 also increases the activity of the MDR1 promoter.
Literature	GENERIF_SUMMARY	E2F1 directly binds to CTSL2 promoter and regulates CTSL2 gene expression.
Literature	GENERIF_SUMMARY	GCN5 has a role in potentiating growth of non-small cell lung cancer by promoting E2F1, cyclin D1, and cyclin E1 expression
Literature	GENERIF_SUMMARY	where miR-17-5p showed a significant down-regulation in 23 non-metastatic HCC biopsies compared to 10 healthy tissues; however, E2F-1 and c-MYC transcripts were markedly elevated
Literature	GENERIF_SUMMARY	MLL5 can associate with HCF-1 and then be recruited to E2F1-responsive promoters to stimulate H3K4 trimethylation and transcriptional activation.
Literature	GENERIF_SUMMARY	EZH2-mediated signal loop, EZH2-CDK4/6-pRb-E2F1, is probably involved in GBM tumorigenicity, and this loop can be blocked by miRNA-138
Literature	GENERIF_SUMMARY	Regulation of E2F1 by the von Hippel-Lindau tumour suppressor protein predicts survival in renal cell cancer patients.
Literature	GENERIF_SUMMARY	Data indicate that PIM-2 is an upstream activator of E2F-1 and ATM in the ultraviolet damage response.
Literature	GENERIF_SUMMARY	SKP2 serves as dual ERa E3 ligase/coactivator for late-activated target genes, revealing a novel mechanism whereby ERa/SCFSKP2 transactivation of E2F-1 feeds forward to drive G1-to-S.
Literature	GENERIF_SUMMARY	E2F-1, a key transcription factor necessary for cell growth, DNA repair, and differentiation, is an attractive target for development of anticancer drugs in tumors that are E2F ""oncogene addicted"""."
Literature	GENERIF_SUMMARY	Experiments demonstrate a role of E2F1 in osteoblast differentiation and mineralization and suggest that FZD1 is required, in part, for E2F1 regulation of

Main Category	Category	Terms
		osteoblast mineralization.
Literature	GENERIF_SUMMARY	UHRF2 is a transcriptional target of E2F, that it directly interacts with E2F1.
Literature	GENERIF_SUMMARY	MAGE-A11 is a proto-oncogene whose increased expression in prostate cancer reverses retinoblastoma-related protein p107 from a transcriptional repressor to a transcriptional activator of the androgen receptor and E2F1.
Literature	GENERIF_SUMMARY	E2F7 recruits the co-repressor C-terminal-binding protein (CtBP) and that CtBP2 is essential for E2F7 to repress E2F1 transcription.
Literature	GENERIF_SUMMARY	MiR-329 may inhibit cell proliferation in human glioma cells through regulating E2F1-mediated suppression of Akt pathway.
Literature	GENERIF_SUMMARY	somatic mutations in DP-1 uncouple normal control of the E2F pathway, and thus define a new mechanism that could contribute to aberrant proliferation in tumor cells
Literature	GENERIF_SUMMARY	Api5, even if not physically interacting with E2F1, contributes positively to E2F1 transcriptional activity by increasing E2F1 binding to its target promoters.
Literature	GENERIF_SUMMARY	report that the transcription factor E2F1 binds only to the E2F1 distal site of the p53 promoter in the human papillomavirus positive carcinoma HeLa cell line
Literature	GENERIF_SUMMARY	Low E2F1 expression is associated with breast and lung neoplasms.
Literature	GENERIF_SUMMARY	Genomic deregulation of the E2F/Rb pathway leads to activation of the oncogene EZH2 in small cell lung cancer.
Literature	GENERIF_SUMMARY	Ei24 is a novel E2F1 target gene contributing to the survival of p53-deficient cells upon UVC irradiation and thus may have a potential significance as a therapeutic target of certain chemotherapy for treating p53-deficient tumors.
Literature	GENERIF_SUMMARY	E2F1's function in the metastatic process in clear cell renal cell carcinoma
Literature	GENERIF_SUMMARY	The results of in vitro and in vivo experiments demonstrated that combined therapy with Ad-E2F and CIK cells produce stronger antitumor effects than the administration of Ad-E2F or CIK cells alone
Literature	GENERIF_SUMMARY	E2F-1 is involved in regulating multiple signaling pathways in reversing multidrug resistance in gastric cancer cells.
Literature	GENERIF_SUMMARY	The E2F1/DNMT1 axis is associated with the development of AR negative castration resistant prostate cancer.
Literature	GENERIF_SUMMARY	PRMT1/PRMT5 arginine methylation regulate the

Main Category	Category	Terms
		biological activity of E2F1 promoting apoptosis/ proliferation.
Literature	GENERIF_SUMMARY	The model of p53 action on E2F1-activated PLK1 gene may explain at least partly how p53 as a suppressor regulates the downstream effects of E2F1 in cellular stresses including DNA damage stress
Literature	GENERIF_SUMMARY	Authors find that phosphorylation of S788 and S795 weakens the direct association between the N-terminal portion of the retinoblastoma protein C-terminal domain and the marked-box domains of E2F.
Literature	GENERIF_SUMMARY	Analysis data from a panel of cell cycle transcription factors (E2F1, E2F4, E2F6, and GABPA) finds that a set of core cell cycle genes regulated in both U2OS and HeLa cells are bound by multiple cell cycle transcription factors.
Literature	GENERIF_SUMMARY	A novel function of pVHL and demonstrate a negative-feedback loop between pVHL and E2F1, which may shed new light on the explanation of the role of pVHL in tumour suppression.
Literature	GENERIF_SUMMARY	These results demonstrate that E2F1 mRNA is a physiological target of TTP and suggests that TTP controls proliferation as well as migration and invasion through the regulation of E2F1 mRNA stability.
Literature	GENERIF_SUMMARY	ERIC as a novel lncRNA that is transcriptionally regulated by E2F1 transcription factor. [XLOC_006942]
Literature	GENERIF_SUMMARY	E2F1 has a role in chemotherapeutic drug resistance via ABCG2
Literature	GENERIF_SUMMARY	E2F1/Rb pathways are altered by HHV-6, which then uses the E2F1 transcription factor to express viral genes
Literature	GENERIF_SUMMARY	miR202 is activated by E2F1 and in turn downregulates MYCN protein expression in the neuroblastoma cell line LAN5. Upregulation of miR202 may therefore be a novel strategy for neuroblastoma treatment.
Literature	GENERIF_SUMMARY	A differentially regulated gene module, which is enriched for cell cycle-related and E2F1 targeted genes, plays a role in the molecular differences between smoker and non-smoker lung cancer.
Literature	GENERIF_SUMMARY	eleven-nineteen lysine-rich leukemia protein enhanced E2F1 deacetylation via recruitment of histone deacetylase 1
Literature	GENERIF_SUMMARY	Higher expression of e2f-1 occurred in the early stages of gastric adenocarcinoma, with lower expression of it in the later stages.
Literature	GENERIF_SUMMARY	E2F1 was expressed in coordination with TFDP3 in most prostate cancer tissues.
Literature	GENERIF_SUMMARY	E2F-dependent expression of DSCC1 confers anti-apoptotic properties in colorectal cancer cells.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Hypermethylated E2F transcription factor 1 (E2F1) motif is a key regulatory element for the DNMT1 gene in BRCA1-mutated breast cancer
Literature	GENERIF_SUMMARY	we identify the contribution of dysregulated C/EBPalpha and E2F1 to elevated Trib2 expression and leukemic cell survival
Literature	GENERIF_SUMMARY	These results suggested that Rta represses IRF-3 expression by increasing E2F1 binding to the IRF-3 promoter.
Literature	GENERIF_SUMMARY	E2F1 is a potent biomarker, as well as a therapeutic target for all-trans retinoic acid -based differentiation therapeutics, and raise the hope of using differentiation-based approaches for osteosarcoma patients.
Literature	GENERIF_SUMMARY	E2F1 activated Egr-1 expression in a biphasic fashion as described in both human and mouse hepatocytes
Literature	GENERIF_SUMMARY	Studied a new therapeutic alternative in the treatment of multiresistant lung adenocarcinoma via siRNA-specific transfection of six crucial molecules involved in lung carcinogenesis
Literature	GENERIF_SUMMARY	Accumulation of E2F1 in the nucleolus is an indicator for nucleolar stress, which buffers elevation of E2F1 in the nucleoplasm and coordinates the diversifying mechanisms of E2F1 acts in cell cycle progression and apoptosis in response to DNA damage.
Literature	GENERIF_SUMMARY	HES6 is up-regulated in aggressive human prostate cancer and drives tumour growth by enhancing the transcriptional activity of the AR, which is preferentially directed to a regulatory network enriched for transcription factors such as E2F1.
Literature	GENERIF_SUMMARY	E2F1 knockdown inhibited prostate tumor growth in vitro and in vivo through sensitizing tumor cells to ICAM-1 mediated anti-immunity by NF-kappaB modulation, highlighting the potential of E2F1 as a therapeutic target.
Literature	GENERIF_SUMMARY	E2F1 acts as a transcriptional activator for MMPs and directly enhances MMP transcription by binding to E2F1 binding sequences in the promoter, or indirectly activates MMPs through enhanced Sp1 and NF-kappa B as a consequence of E2F1 activation in SCLC.
Literature	GENERIF_SUMMARY	Over-expression of E2F1 is associated with hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	Our findings suggest that EZH2-specific microRNA-98 can effectively inhibit cell proliferation in vitro and regulate the pRb-E2F pathway in human ovarian cancer
Literature	GENERIF_SUMMARY	The target of retinoblastoma protein, E2F1, is a transcription factor whose expression is increased in many human cancers . [Review]
Literature	GENERIF_SUMMARY	Data show that E2F-1 form a complex with RNA polymerase II and protein PURA for transcriptional

Main Category	Category	Terms
		activation of the secondary promoter.
Literature	GENERIF_SUMMARY	MiR-320 regulates the expression and functions of E2F1 and SF-1 in Polycystic Ovary Syndrome patients.
Literature	GENERIF_SUMMARY	these data show a synergistic effect of MDM2 and E2F1 on TAp73 transcriptional activity, suggesting a novel regulation pathway of TAp73.
Literature	GENERIF_SUMMARY	E2F1 is a novel modulator of MDM2 expression via SNP309, as well as mediating cell proliferation and CCND1 expression in skin cancer.
Literature	GENERIF_SUMMARY	HBXIP facilitates the proliferation of hepatoma cells by up-regulating SCG3 via E2F1 and miR-509-3p modulation.
Literature	GENERIF_SUMMARY	These results reveal a novel link between ROR-alpha and E2F1 in regulating cell cycle progression and mammary tissue morphogenesis.
Literature	GENERIF_SUMMARY	These results suggest that the ARF promoter, which is specifically activated by deregulated E2F activity, is an excellent candidate to drive therapeutic cytotoxic gene expression, specifically in cancer cells.
Literature	GENERIF_SUMMARY	The involvement of E2F1-dependent DROSHA activation in pri-miRNA processing under DNA damage stress will provide further insight into the regulation of miRNA biosynthesis.
Literature	GENERIF_SUMMARY	Data indicate drug-induced ligands gene expression was transcriptionally regulated and involved DNA damage response (DDR)-dependent E2F1 transcription factor activity.
Literature	GENERIF_SUMMARY	Data indicate that a positive correlation was found between expression of transcription factor E2F-1 and tumor supressor p53 in Hodgkin lymphoma (HL).
Literature	GENERIF_SUMMARY	Results showed that E2F1 acts as a negative feedback regulator of c-Myc-induced hTERT transcription. If this negative regulating mechanism of E2F1 is attenuated or abrogated, hTERT is easily activated by c-Myc overexpression, leading to tumorigenesis.
Literature	GENERIF_SUMMARY	a regulatory role for KDM2A in breast cancer cell invasion and migration, through the regulation of E2F1 function
Literature	GENERIF_SUMMARY	stimulation of the E2F1-RHAMM axis in aggressive cancer cells is of high clinical significance
Literature	GENERIF_SUMMARY	Set7/9 is a potential biomarker in tumour cells and is associated with overexpressed E2F1 activity.
Literature	GENERIF_SUMMARY	Identification of the transcription factor binding site of the HOXB9 promoter region suggests that E2F1 is a direct regulator of HOXB9 expression in breast cancer.
Literature	GENERIF_SUMMARY	High E2F1 expression is associated with cisplatin resistance and glioma.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ATO has potent in vitro and in vivo activity in lung adenocarcinoma, partially mediated through E2F1 downregulation and apoptosis
Literature	GENERIF_SUMMARY	Myc is critical in modulating the amplitude of cell cycle, whereas cyclin D/E activities have little effect on amplitude but do contribute to the modulation of duration of E2F1 activation, thereby affecting the pace of cell cycle progression.
Literature	GENERIF_SUMMARY	Our findings may be helpful in exploring the underlying mechanisms of CML and provide useful information in cancer treatment.
Literature	GENERIF_SUMMARY	cancer-related miR-203 is a novel transcriptional target of E2F1 and that it regulates cell cycle arrest by participating in a feedback loop with E2F1
Literature	GENERIF_SUMMARY	The release of BIN1 from hypo-poly(ADP-ribosyl)ated E2F1 is a mechanism by which serum starvation promotes E2F1-induced apoptosis.
Literature	GENERIF_SUMMARY	Calpain cleavage of E2F1 was found to be in response to NMDA receptor HIV-mediated neurotoxicity.
Literature	GENERIF_SUMMARY	A newly discovered role of E2F1 in the regulation of p53R2 expression in DNA damage response.
Literature	GENERIF_SUMMARY	Data identify the metastasis suppressor TXNIP as new target of miR-224/miR-452 that induces feedback inhibition of E2F1 and show that miR-224/452-mediated downregulation of TXNIP is essential for E2F1-induced EMT and invasion
Literature	GENERIF_SUMMARY	E2F1 regulates cellular senescence by inhibiting FOXO3.
Literature	GENERIF_SUMMARY	PRMT4 has a role in arginine methylation, which negatively regulates retinoblastoma tumor suppressor protein and promotes E2F-1 dissociation
Literature	GENERIF_SUMMARY	excess E2F1 due to Rb inactivation recruits the complex of Cdc20 and the anaphase-promoting complex/ cyclosome to deregulate the expression of UBCH10
Literature	GENERIF_SUMMARY	High E2F1 expression is associated with metastasis in non-small cell lung cancers.
Literature	GENERIF_SUMMARY	THAP11, ZNF143, and HCF-1 form a mutually dependent complex on chromatin, which is independent of E2F occupancy.
Literature	GENERIF_SUMMARY	E2F1 microduplications or microdeletions are present in men with NOA (7.3%).
Literature	GENERIF_SUMMARY	High E2F1 expression is associated with response to low intravenous immunoglobulin therapy in patients with Kawasaki disease.
Literature	GENERIF_SUMMARY	E2F-1 inhibits growth of gastric cancer cells by regulating multiple signaling pathways.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Stimulation of multiplenon-small cell lung cancer cell lines with nicotine led to enhanced recruitment of beta-arrestin-1 and E2F1 on vimentin
Literature	GENERIF_SUMMARY	Tudor-SN is a potential substrate of G1/S phase Cyclin-Dependent Kinases, and promotes cell cycle progression by facilitating E2F-1-mediated gene transcription.
Literature	GENERIF_SUMMARY	this novel function of E2F may have a major impact on cell viability, and it is independent of induction of apoptotic genes. Here, we discuss the implications of these findings in cancer biology.
Literature	GENERIF_SUMMARY	Inhibition of miR-10b represses E2F1-mediated transcription, leading to down-regulation of multiple E2F1 target genes and stalling progression through the S-phase of cell cycle.
Literature	GENERIF_SUMMARY	the functional involvement of the Pin1-E2F1 axis in the anti-proliferative effect of BZB indicating Pin1-E2F as an attractive target to control HCC cell growth.
Literature	GENERIF_SUMMARY	Results show that in triple negative breast cancer cells, PELP1 interacts with mutated Tp53 and E2F1 to differentially regulates their target gene expression.
Literature	GENERIF_SUMMARY	Indicate a novel oncogenic role of E2F1 by enhancing glycolysis via suppression Sirt6 transcription in bladder and prostate tumor cell lines.
Literature	GENERIF_SUMMARY	E2F1 accumulates at sites of oxidative and UV-induced DNA damage, and interact with gammaH2AX DNA repair factor.
Literature	GENERIF_SUMMARY	Protein expression of E2F1, a cell-cycle regulator, is negatively regulated by miR-320.
Literature	GENERIF_SUMMARY	NF-YB transcription factor was identified as a novel direct E2F1 target.
Literature	GENERIF_SUMMARY	Understanding the role of E2F1 in activating RRM2 transcription will help to explain the relationship between E2F1 and RRM2 in CRC
Literature	GENERIF_SUMMARY	An LXCXE motif in UL97 helps activate E2F-dependent promoters independently of its stimulatory effect on Rb phosphorylation.
Literature	GENERIF_SUMMARY	E2F1 and E2F3 are present within the syncytiotrophoblast of placenta and that E2F1 is reduced in preeclampsia. Although silencing of either E2F1 or E2F3 does not alter MMP14 expression, both appear to regulate soluble endoglin release.
Literature	GENERIF_SUMMARY	these data reveal a novel mechanism for the regulation of the expression of HIF2a, demonstrating that the HIF2a promoter is regulated by E2F1 directly and that Cezanne regulates HIF2a expression through control of E2F1 levels
Literature	GENERIF_SUMMARY	E2F1 transcripts and miR-17-5p were significantly

Main Category	Category	Terms
		downregulated while c-Myc and MxA transcripts were significantly upregulated in SLE.
Literature	GENERIF_SUMMARY	the BIM gene was shown to be activated via atypical E2F-responsive promoter elements and to contribute to E2F1-mediated apoptosis.
Literature	GENERIF_SUMMARY	Data suggest that KDM1A is overexpressed in oral cancer and promotes its proliferation via E2F1 signaling and confers poor clinical outcome.
Literature	GENERIF_SUMMARY	gene amplification, p53 and Rb disruption, and E2F1 activation drive FOXM1 expression in epithelial ovarian cancer, and FOXM1 promotes cell cycle progression in cell models
Literature	GENERIF_SUMMARY	Activation of EZH2 and SUZ12 Regulated by E2F1 Predicts the Disease Progression and Aggressive Characteristics of Bladder Cancer
Literature	GENERIF_SUMMARY	In this review, we discuss the role of the ubiquitin-proteasome system in proteolytic control of RB-E2F pathway components, and recent data that points to surprising non-proteolytic roles for the ubiquitin-proteasome system
Literature	GENERIF_SUMMARY	Epigenetic factor EPC1 is a master regulator of DNA damage response by interacting with E2F1 to silence death and activate metastasis-related gene signatures
Literature	GENERIF_SUMMARY	cell cycle-dependent transcription of the TRAIP gene by E2F1, E2F2, and E2F4 and rapid protein degradation leads to cell cycle-dependent expression with a maximum in G2/M
Literature	GENERIF_SUMMARY	These results uncover a novel mechanism for E2F1 transcriptional activation through removal of its Lys-63-linked ubiquitination by UCH37.
Literature	GENERIF_SUMMARY	mTORC2 promotes cell survival by stimulating E2F1 expression through a c-Myc- and miR-9-3p-dependent mechanism.
Literature	GENERIF_SUMMARY	MageB2 counteracts E2F inhibition by ribosomal proteins independently of Mdm2 expression
Literature	GENERIF_SUMMARY	Our findings define an intricate E2F1-dependent mechanism by which Id1 increases thymidylate synthase and IGF2 expressions to promote cancer chemoresistance. The Id1-E2F1-IGF2 regulatory axis has important implications for cancer prognosis and treatment.
Literature	GENERIF_SUMMARY	miR-342-3p is capable of indirectly regulating MYC activity via direct repression of E2F1.
Literature	GENERIF_SUMMARY	the present findings support the conclusion that the E2F1 transcription factor regulates the expression of ADAM12 by binding differential cis-acting elements.
Literature	GENERIF_SUMMARY	Deubiquitylase POH1 stabilizes E2F1 protein through binding to and deubiquitylating E2F1 in liver cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	By targeting genes involved in controlling pRb/E2F1 activity, miR-449a regulates cell cycle progression and apoptosis and consequently enhances the radiosensitivity of PC-3 cells. Thus, miR-449a, as a miRNA component of the Rb pathway, promotes the radiosensitivity of PC-3 cells through regulating pRb/E2F1.
Literature	GENERIF_SUMMARY	E2F1 negatively regulates both Ogt and Mgea5 expression in an Rb1 protein-dependent manner.
Literature	GENERIF_SUMMARY	Data show that long non-coding RNA HULC promotes tumor angiogenesis in liver cancer through microRNA miR-107/E2F1 transcription factor (E2F1)/sphingosine kinase 1 (SPHK1) signaling.
Literature	GENERIF_SUMMARY	In this review, the general structure and function of E2Fs as well as the role for E2Fs in the development of HCC and liver disease is evaluated.
Literature	GENERIF_SUMMARY	Data suggest that the SNP rs2853669 variant induces telomerase reverse transcriptase (TERT) transcription levels by blocking E2F1 transcription factor (E2F1) binding to its promoter.
Literature	GENERIF_SUMMARY	Results reveal an lncRNA- and E2F1-driven regulatory loop in which E2F1-dependent induction of antisense RNA leads to changes in chromatin structure, facilitating E2F1-dependent expression of SPHK1 and restriction of E2F1-induced apoptosis.
Literature	GENERIF_SUMMARY	TGFbeta induces autophagy through the pRb/E2F1 pathway and transcriptional activation of autophagy-related genes
Literature	GENERIF_SUMMARY	Upregulation of E2F1 has been implicated in oncogenesis, prognosis, and prediction of therapeutic response but is not seemingly to have a relationship with the accused SV40.
Literature	GENERIF_SUMMARY	demonstrate that E2F1 regulates lipid synthesis and glycolysis and thus contributes to the development of liver pathology
Literature	GENERIF_SUMMARY	E2F1 expression induced human colon cancer cell death, and suppression of GCN5 expression in cells with E2F1 overexpression further facilitated cell apoptosis
Literature	GENERIF_SUMMARY	overexpression of ectopic E2F-1 protein may render cells more sensitive to gemcitabine-mediated apoptosis
Literature	GENERIF_SUMMARY	Results uncovered the collaboration between EZH2 and E2F1 in controlling downstream genes that are highly expressed in aggressive tumors.
Literature	GENERIF_SUMMARY	results suggested the existence of E2F1-miR-34c-SCF negative feedback loop which was interrupted by the hyper-methylation of miR-34c promoter in CRC cells and increased cell proliferation.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	These data suggest that CBX8 modulates SIPS through the RB-E2F1 pathway in CML cells and provide important insight into its application in CML treatment.
Literature	GENERIF_SUMMARY	The combined E2F1/TS immunophenotype could be a potential indicator of colon cancer sensitivity to 5FU.
Literature	GENERIF_SUMMARY	HER2 signaling promotes breast cancer cell proliferation through regulation of E2F1-driven DNA metabolism.
Literature	GENERIF_SUMMARY	Integrative genomic analyses confirm that E2F4 or E2F1 expression level is high in liposarcoma patients which associate with unfavorable prognosis.
Literature	GENERIF_SUMMARY	Chk1's expression is controlled by p53 and RB/E2F1 at the transcriptional level.
Literature	GENERIF_SUMMARY	E2F1 couples immune cell development to immune response.E2f1 role in the inflammation-associated cancers [review]
Literature	GENERIF_SUMMARY	Study found that E2F1 is overexpression in prostate cancer (PCa) tissues and significantly associated with high Gleason score. These results suggest that E2F1 may promote tumor cell invasion and migration through the regulation of CD147 in PCa cells.
Literature	GENERIF_SUMMARY	We suggest that MNNG-stimulated ATR/CHK1 signaling stabilizes E2F3 by S124 phosphorylation, and then E2F3 together with NFY co-transactivate RRM2 expression for DNA repair.
Literature	GENERIF_SUMMARY	Specific E2Fs also have prognostic value in breast cancer, independent of clinical parameters. We discuss here recent advances in understanding of the RB-E2F pathway in breast cancer. We also discuss the application of genome-wide genetic screening efforts to gain insight into synthetic lethal interactions of CDK4/6 inhibitors in breast cancer for the development of more effective combination therapies.
Literature	GENERIF_SUMMARY	AR activity was inversely associated with cell proliferation, whereas the expression of Fa complex genes was correlated with elevated cell cycle progression, expression of the E2F transcription factor 1 (E2F1) and loss of RB1.
Literature	GENERIF_SUMMARY	This review focuses on the relationship between E2F1, growth factors and cytokines.
Literature	GENERIF_SUMMARY	High E2F1 expression is associated with glioma.
Literature	GENERIF_SUMMARY	We conclude that high expression of S18-2 and free E2F...1might be a good set of prognostic markers for endometrial cancer
Literature	GENERIF_SUMMARY	transcription-based mechanism that mediates the inflammatory effect of PAD4 and establishes the interplay between citrullination and acetylation in the control of E2F-1 as a regulatory interface for driving inflammatory gene expression

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Data suggest that the protein arginine methyltransferase 5 (PRMT5)-E2F1 transcription factor (E2F-1) pathway may act as a common target for exogenous lectins including Anguilla japonica lectin 1 (AJL1), and the cellular response to exogenous AJL1 may suggest a novel agent for cancer gene therapy.
Literature	GENERIF_SUMMARY	Expression of Kv10.1 driven by phosphorylated Rb/E2F1 contributes to G2/M progression of cancer and non-transformed cells.
Literature	GENERIF_SUMMARY	High E2F1 expression is associated with gastric cancer.
Literature	GENERIF_SUMMARY	the E2F1-mediated electron transport chain-dependent mechanism has emerged as the regulatory mechanism of cell cycle progression in breast cancer cells
Literature	GENERIF_SUMMARY	REVIEW
Literature	GENERIF_SUMMARY	Results suggest that E7 recruited CUL2, driven by CUL2/E2F1/miR-424 regulatory loop, is overexpressed and accelerates HPV16-induced cervical carcinogenesis.
Literature	GENERIF_SUMMARY	These data suggest a model in which cells experiencing oncogene-induced replication stress through deregulation of E2F-dependent transcription.
Literature	GENERIF_SUMMARY	our results indicate that E2F1 is an important downstream gene of ISX in hepatoma progression.
Literature	GENERIF_SUMMARY	Using iterative experimental and computational analyses, the authors show physical and functional interactions between NF-kappaB and the E2 Factor 1 (E2F-1) and E2 Factor 4 (E2F-4) cell cycle regulators.
Literature	GENERIF_SUMMARY	nicotine-mediated induction of alpha7 nAChR was reduced when E2F1 was depleted and in contrast elevated when STAT1 was depleted by siRNAs.
Literature	GENERIF_SUMMARY	Suppression of tumorigenesis by CADM1 may be mediated by the Rb-E2F1 pathway.
Literature	GENERIF_SUMMARY	E2F1 directly facilitated BMCC1 transcription. These results also suggest that BMCC1 induced by E2F1 acts as a tumor suppressor through its pro-apoptotic function, resulted in favorable prognosis of neuroblastoma.
Literature	GENERIF_SUMMARY	E2F1 was significantly increased during Oral Tongue Carcinoma Metastasis.
Literature	GENERIF_SUMMARY	E2F1 was confirmed as a direct target of miR-622.
Literature	GENERIF_SUMMARY	E2F1 forms a protein complex with EBNA3C and E2F6, and EBNA3C competes with E2F1 for E2F6 binding. E2F6 is also recruited by EBNA3C to the E2F1 promoter, which is critical for EBNA3C-mediated cell proliferation.
Literature	GENERIF_SUMMARY	H19 is overexpressed and plays oncogenic role in PDAC through promoting cancer cell proliferation via the upregulation of E2F-1.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The pyruvate dehydrogenase kinases (PDKs) PDK1 and PDK3 are direct targets of KDM4A and E2F1 and modulate the switch between glycolytic metabolism and mitochondrial oxidation.
Literature	GENERIF_SUMMARY	In NSCLC patients, E2F1 expression was significantly correlated with age, sex, and tumor stage. E2F1 and E2F2 overexpression showed a significant association with poor prognosis.
Literature	GENERIF_SUMMARY	Results provide evidence that E2F1rs3213180 polymorphism may influence susceptibility to HPV-associated oral squamous cell carcinoma. particularly oropharynx tumors.
Literature	GENERIF_SUMMARY	Higher levels of miR-135a in gastric cancer (GC) are associated with shorter survival times and reduced times to disease recurrence. The mechanism whereby miR-135a promotes GC pathogenesis appears to be the suppression of E2F1 expression.
Literature	GENERIF_SUMMARY	These data suggested that miR-320 regulates the proliferation and cell cycle by targeting E2F1 in human OsC progression.
Literature	GENERIF_SUMMARY	our results established the significance of activating the KLF6-E2F1 axis in aggressive primary metastatic clear cell renal cell carcinoma
Literature	GENERIF_SUMMARY	role for E2F1 and TFDP1 in the transcriptional regulation of PITX1 in articular chondrocytes
Literature	GENERIF_SUMMARY	Here, we introduce a transcriptional signature to specifically track CDK2 activity. It responds to genetic and chemical perturbations in the CDK-RB-E2F axis, correlates with mitotic rate in vitro and in vivo and reacts rapidly to changes in CDK2 activity during cell cycle progression
Literature	GENERIF_SUMMARY	these results suggest that the E2F1/miR19a/PPARalpha feedback loop is critical for glioma progression
Literature	GENERIF_SUMMARY	COMMD9 participates in TFDP1/E2F1 activation and plays a critical role in non-small cell lung cancer.
Literature	GENERIF_SUMMARY	activation of the PI3K pathway does not suppress activation of the ARF or BIM gene by over-expressed E2F1
Literature	GENERIF_SUMMARY	Results provide mechanistic insight into a series of complex, differentiation-specific molecular mechanisms that regulate E2F1 during keratinocyte maturation via multiple events. These events include nucleocytoplasmic transport and changes in ubiquitinylation patterns specifically orchestrated through S403 and T433, and which differ from other mechanisms that regulate E2F1 turnover in undifferentiated cells.
Literature	GENERIF_SUMMARY	E2F1-mediated hPOMC transcription is a potential target for suppressing ACTH production in ectopic

Main Category	Category	Terms
		Cushing's syndrome.
Literature	GENERIF_SUMMARY	E2F1 is significantly up-regulated in glioblastoma.
Literature	GENERIF_SUMMARY	inhibition of PDK4 activity in Hepatocellular carcinoma cells increased cyclin E1, cyclin A2, and E2F1 proteins.
Literature	GENERIF_SUMMARY	Taken together, this study reveals evidence demonstrating a mechanism by which the LPR6/GSK3beta/E2F1 axis-upregulated LSH promoted gliomas.
Literature	GENERIF_SUMMARY	High E2F1 expression is associated with melanoma.
Literature	GENERIF_SUMMARY	This study suggests for the first time an involvement of E2F1 copy number variations in testicular germ cell tumor susceptibility and supports previous preliminary data on the importance of AKT/mTOR signaling pathway in this cancer.
Literature	GENERIF_SUMMARY	E2F transcription factor 1 (E2F1) downregulated micrRNA miR-519d directly and miR-519d downregulated Ras homolog gene family member C (RhoC) directly.
Literature	GENERIF_SUMMARY	High E2F1 expression is associated with decreased radiosensitivity in nasopharyngeal carcinoma .
Literature	GENERIF_SUMMARY	Results demonstrated that E2F-1 mediated PEG10 overexpression promotes pancreatic cancer cell proliferation via accelerating cell cycle progression and increase migration and invasion through ERK/MMP7 pathway.
Literature	GENERIF_SUMMARY	The low level E2F1 are sufficient to induce numerous cell cycle-promoting genes, intermediate levels induce growth arrest genes (i.e., p18, p19 and p27), whereas higher levels are necessary to induce key apoptotic E2F1 targets APAF1, PUMA, HRK and BIM.
Literature	GENERIF_SUMMARY	This study reveals a molecular pathway involving lncRNA GAS5/E2F1/P27(Kip1) which regulates cell proliferation and could be a potential therapeutic target in prostate cancer.
Literature	GENERIF_SUMMARY	Data show that lncRNA-HIT acted as an oncogene through association with E2F transcription factor 1 protein (E2F1).
Literature	GENERIF_SUMMARY	Propose that E2F1 interacts with BRCA1 indirect pathway to induce two different small molecule metabolic pathways or cell cycle regulation pathways in hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	Cell proliferation and apoptosis were almost completely abolished in the PAa cells cotreated with TRIM28 siRNA and etoposide following knockdown of E2F1. The results of our study demonstrated that the combination of TRIM28 siRNA and etoposide may be effective against nonsmall cell lung cancer (NSCLC)and has the potential of being a new therapeutic tool for future treatment.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Heavy ion irradiation could induce p53(-/-) hepatoma cells to undergo apoptosis via E2F1/Bax/Casp3 signaling pathway.
Literature	GENERIF_SUMMARY	These results reveal an additional level of regulation of the stability and the activity of E2F1 by a non-degradative K63-poly-ubiquitination and uncover a novel function for the E3-ubiquitin ligase cIAP1.
Literature	GENERIF_SUMMARY	E2F1 induces TINCR transcriptional activity and accelerates gastric cancer progression via activation of TINCR/STAU1/CDKN2B signaling axis.
Literature	GENERIF_SUMMARY	Describe a regulatory loop miR-218-CDK6/CyclinD1-E2F1 whose disruption may contribute to cell cycle progression in gastric cancer.
Literature	GENERIF_SUMMARY	ANKRD22 exhibits oncogene activity that promotes tumor progression in NSCLC through the transcriptional regulation of E2F1.
Literature	GENERIF_SUMMARY	Results demonstrated that somatic mutation within E2F1
Literature	GENERIF_SUMMARY	p63alpha protein up-regulates heat shock protein 70 expression via E2F1 transcription factor 1, promoting Wasf3/Wave3/MMP9 signaling and bladder cancer invasion
Literature	GENERIF_SUMMARY	These results demonstrate that gambogic acid sensitizes pancreatic cancer cells to gemcitabine in vitro and in vivo by inhibiting the activation of the ERK/E2F1/RRM2 signaling pathway.
Literature	GENERIF_SUMMARY	By comparison in HeLa, Dox induced apoptosis through upregulation of endogenous E2F1 involving post-transcriptional mechanisms, while E2F6 was down regulated with induction of the Checkpoint kinase-1 and proteasome degradation. These data imply that E2F6 serves to modulate E2F activity and protect cells including cardiomyocytes from apoptosis and improve survival.
Literature	GENERIF_SUMMARY	that E2F1 mRNA stability and E2F1 protein levels are reduced in cells lacking RALY expression
Literature	GENERIF_SUMMARY	USP11 depletion suppresses cell proliferation and wound healing in lung epithelial cells, and these effects are reversed by E2F1 and PEG10 overexpression.
Literature	GENERIF_SUMMARY	E2F1 knockdown decreased the expression of discoidin domain receptor 1 (DDR1) which plays a crucial role in many fundamental processes such as cell differentiation, adhesion, migration and invasion.
Literature	GENERIF_SUMMARY	Our data imply that downregulation of E2F1 may be a key factor in the celastrol-mediated inhibitory effects in HepG2 cells, and celastrol can serve as a leading compound for the development of compounds designed to inactivate E2F1 for hepatocellular carcinoma therapy

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	E2F1 enhances the invasion and metastasis of prostate cancer through a variety of mechanisms, and its expression level has an important relationship with the adverse prognosis of patients with prostate cancer
Literature	GENERIF_SUMMARY	PTEN-4A preferentially localized to the nucleus where it suppressed E2F1-mediated transcription of cell cycle genes.
Literature	GENERIF_SUMMARY	Reports showed that high expression of the transcription factor E2F1 was involved in the invasion and metastasis of small cell lung cancer (SCLC). Further results provided evidence that E2F1 promoted EMT by regulating ZEB2 gene expression in SCLC.
Literature	GENERIF_SUMMARY	CacyBP expression is regulated by E2F1, EGR1, and CREB transcription factors in colorectal cancer HCT116 cells.
Literature	GENERIF_SUMMARY	identification of protumorigenic transcriptional networks specific to RB loss that were validated in clinical samples demonstrated the ability of RB loss to differentially reprogram E2F1 in human cancers.
Literature	GENERIF_SUMMARY	The nuclear transcription factor Y subunit beta (NFYB)-E2F transcription factor 1 (E2F1) pathway displays a crucial role in the chemoresistance ofoxaliplatin-resistant colorectal cancer (OR-CRC) by inducing the expression and activation of checkpoint kinase 1 (CHK1), suggesting a possible therapeutic target for oxaliplatin resistance in CRC.
Literature	GENERIF_SUMMARY	found that human E2F1 competes with YAP for TEAD1 binding, affecting YAP activity, indicating that this mode of cross-regulation is conserved
Literature	GENERIF_SUMMARY	E2F1 interacts with BCL-xL independently from its BH3 binding interface and induces a stabilization of BCL-xL at mitochondrial membranes. This prevents efficient control of BCL-xL over its binding partners, in particular over BAK resulting in the induction of cell death.
Literature	GENERIF_SUMMARY	Data show that mRNA translation stress induces E2F transcription factor 1 (E2F1) via PI3-kinase p110 subunit delta (PI3Kdelta).
Literature	GENERIF_SUMMARY	the overexpression of TACC3 or E2F1 indicates a high sensitivity to cisplatin.
Literature	GENERIF_SUMMARY	inhibition of hsa-mir183 reduced proliferation and impaired colony formation of CML SPCs. Downstream of this, inhibition of E2F1 also reduced proliferation of CML SPCs, leading to p53-mediated apoptosis. In addition, we demonstrate that E2F1 plays a pivotal role in regulating CML SPC proliferation status.
Literature	GENERIF_SUMMARY	The long non-coding RNA FOXD2-AS1 promotes bladder cancer progression and recurrence through a positive feedback loop with Akt and E2F1.
Literature	GENERIF_SUMMARY	The network model simulations and experimental data indicate the ability of enhanced expression of both

Main Category	Category	Terms
		miR-205-5p and miR-342-3p to decrease tumor chemoresistance by cooperatively repressing E2F1.
Literature	GENERIF_SUMMARY	ANP32E is an efficient prognostic marker, and it promotes the G1/S transition and induces tumorigenesis of Triple-negative breast cancer cells by transcriptionally inducing E2F1.
Literature	GENERIF_SUMMARY	PPM1B plays a negative role in the activation of the p38-RB1-E2F1 pathway and that targeting PPM1B could be useful in certain types of cancer by stimulating chemotherapy-induced cell death.
Literature	GENERIF_SUMMARY	If, as expected, the consequences of the deregulation of the CDKN1C-E2F1-TP53 axis were the same as those experimentally demonstrated in mouse models, the disruption of this axis might be useful to predict tumor aggressiveness, and to provide the basis towards the development of potential therapeutic strategies in human Precursor T-cell lymphoblastic lymphomas
Literature	GENERIF_SUMMARY	SNHG16 promotes glioma tumorigenesis by sponging miR-20a-5p, leading to the enhancement of its endogenous targets E2F1
Literature	GENERIF_SUMMARY	E2F1 promotes the transcription of lncRNA RGMB-AS1 in papillary thyroid carcinoma (PTC).
Literature	GENERIF_SUMMARY	blocking E2F1 also induced death of melanoma cells resistant to BRAF inhibitors. In conclusion, our studies suggest that targeting the E2F1 signaling pathway may be therapeutically relevant for melanoma.
Literature	GENERIF_SUMMARY	Studied impact of BET proteins and E2F transcription factor 1 (E2F1) in neoplastic genetic transcription in glioblastoma.
Literature	GENERIF_SUMMARY	the expression of miR175p inhibited high glucoseinduced endothelial cell injury by targeting E2F1.
Literature	GENERIF_SUMMARY	NLK boosts cell proliferation and E2F1 activity and controls the cell cycle switch by releasing HDAC1 from the E2F1 complex, thus priming colorectal cancer progression
Literature	GENERIF_SUMMARY	Antitumor effect of lapatinib and cytotoxic agents by suppression of E2F1 in HER2positive breast cancer.
Literature	GENERIF_SUMMARY	The HBXIP induces PKM2 through transcription factor E2F1 to facilitate ER+ breast cancer cell proliferation.
Literature	GENERIF_SUMMARY	XPC is an RNA polymerase II cofactor recruiting ATAC coactivator complex to promoters by interacting with E2F1.
Literature	GENERIF_SUMMARY	Long non-coding RNA NNT-AS1 sponges miR-424/E2F1 to promote the tumorigenesis and cell cycle progression of gastric cancer.
Literature	GENERIF_SUMMARY	Our findings reveal that the rs3213173 (C/T) and rs3213176 (G/A) polymorphisms of the E2F1 gene are

Main Category	Category	Terms
		genetic risk factors for susceptibility to lung cancer and Head and Neck cancer in the North Indian Population.
Literature	GENERIF_SUMMARY	These results revealed a novel mechanism of E7 function whereby elevated GCN5 acetylates histones and c-Myc to regulate E2F1 expression and cell cycle progression.
Literature	GENERIF_SUMMARY	High E2F1 expression is associated with hepatocellular carcinoma progression.
Literature	GENERIF_SUMMARY	our findings suggest that E2F1-topoII beta signaling may play a role in regulation of cell cycle exit and neuronal differentiation.
Literature	GENERIF_SUMMARY	Study demonstrated that CDCA3 may target p21 to promote colorectal cancer (CRC) cell proliferation and tumorigenesis, at least partially in an E2F1-mediated manner, and that CDCA3 may serve as a potential prognostic and therapeutic target of CRC.
Literature	GENERIF_SUMMARY	KPNA2 mediated nuclear localization of E2F1 and E2F7, where they in turn controlled KPNA2 expression. Taken together, our data provided mechanistic insights into divergently transcriptional regulation of KPNA2, thus pointing to KPNA2 as a potential target for cancer therapy.
Literature	GENERIF_SUMMARY	E2F1 requires cIAP1 for chromatin binding.
Literature	GENERIF_SUMMARY	Data concluded that E2F1-mediated MNX1-AS1-miR-218-5p-SEC61A1 feedback loop contributed to the progression of colon adenocarcinoma.
Literature	GENERIF_SUMMARY	finding supports that DLX6-AS1 accelerates the glioma carcinogenesis by competing endogenous sponging miR-197-5p to relieve E2F1, acting as a novel therapeutic target for glioma.
Literature	GENERIF_SUMMARY	The inverse correlation between XBP1 and E2F1 or MICA expression was corroborated in RNA-Seq analysis.
Literature	GENERIF_SUMMARY	Mechanistic investigation revealed that active PARP-1 served to enhance E2F1 transcription factor activity, and specifically promoted E2F1-mediated induction of DNA repair factors involved in homologous recombination (HR).
Literature	GENERIF_SUMMARY	TopBP1 transcriptional activity is regulated by AKT, and treatment with AKT inhibitors suppresses expression of E2F1 and p73 without interfering with ATR signaling.
Literature	GENERIF_SUMMARY	This study shows a link between E2F1 copy number variations and male infertility
Literature	GENERIF_SUMMARY	Results find that high FASN expression is an adverse prognostic factors for tumor recurrence and progression in non-muscle-invasive bladder cancer (NMIBC) and these patients should be monitored thoroughly for disease recurrence and progression.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The authors predict that the induction of expression of RNASEH2A via human papillomavirus 16 E7 and E2F1 may promote DNA replication and cancer cell proliferation.
Literature	GENERIF_SUMMARY	study unveils an E2F1-specific signaling circuit that constitutively activates ATM and provokes cisplatin resistance in BIN1-deficient cancer cells and further reveals that gammaH2AX emergence may not always reflect DSBs if BIN1 is absent.
Literature	GENERIF_SUMMARY	E2F1 could bind directly to the promoter region of LMCD1-AS1 and activate its transcription.
Literature	GENERIF_SUMMARY	it is found in our study that LncRNA FER1L4 expression is positively correlated with E2F1 mRNA expression. After knockdown of FER1L4 expression, E2F1 expression is significantly down-regulated, whereas the expression of miR-372 is significantly up-regulated; the up-regulation of miR-372 leads to significant down-regulation of FER1L4 and E2F1 expression
Literature	GENERIF_SUMMARY	results indicated that the attenuation of the binding affinity of E2F1 for the POLD1 promoter, mediated by an age-related decline in E2F1 and increased methylation of CpG island 3, downregulates POLD1 expression in aging
Literature	GENERIF_SUMMARY	E2F1 has a cancer protective role in oncogenic BRAF-activated melanoma cells.
Literature	GENERIF_SUMMARY	Cyclin F controls cell-cycle transcriptional outputs by directing the degradation of the E2F1, E2F2, and E2F3a.
Literature	GENERIF_SUMMARY	Environmental insults, such as heat stress, could contribute to an aberrant E2F1 activity by inducing E2F1 mRNA expression. Therefore, subjects with multiple constitutive copies of E2F1 are at greater risk of developing melanoma when exposed to heat. Altogether these results corroborate with the hypothesis that susceptibility to melanoma depends on both the environment and genetic factors.
Literature	GENERIF_SUMMARY	HAGLR could physically interact with DNMT1, and recruit DNMT1 on E2F1 promoter to increase local DNA methylation. HAGLR promoted lung adenocarcinoma progression by recruiting DNMT1 to modulate the promoter methylation and expression of E2F1.
Literature	GENERIF_SUMMARY	E2F1 modulated mitochondrial fusion and mitophagy through regulation of MFN2 in HeLa cells.
Literature	GENERIF_SUMMARY	Inhibition of breast cancer cells by targeting E2F-1 gene and expressing IL15 oncolytic adenovirus.
Literature	GENERIF_SUMMARY	MIR888-dependent inhibition of APLF by E2F1 was demonstrated using overexpression and knockdown experiments, in combination with luciferase assays.
Literature	GENERIF_SUMMARY	High level of DNAJA1 predicted poor prognosis for colorectal cancer (CRC) patients. Its expression was highly linked with E2F1 and CDC45 in CRC tissues.

Main Category	Category	Terms
		More importantly, KNK437 significantly suppressed the growth of DNAJA1 expressing tumor in vivo
Literature	GENERIF_SUMMARY	PIR promotes tumorigenesis of breast cancer by upregulating cell cycle activator E2F1.
Literature	GENERIF_SUMMARY	we investigated the intracellular localization of E2F1, RAD51, and replication protein A (RPA) in colon cancer cells exposed to methyl methanesulfonate (MMS), which induces DNA double-strand breaks (DSBs). In cells experiencing DNA damage, the number of RAD51 and RPA foci increased, suggesting that 3'-single-stranded DNAs were exposed
Literature	GENERIF_SUMMARY	Rb-E2F bistable switch controls cell cycle entry. (Review)
Literature	GENERIF_SUMMARY	LncRNA SNHG3 is activated by E2F1 and promotes proliferation and migration of non-small-cell lung cancer cells through activating TGF-beta pathway and IL-6/JAK2/STAT3 pathway.
Literature	GENERIF_SUMMARY	Ribosomal protein uL3 targets E2F1 and Cyclin D1 in cancer cell response to nucleolar stress.
Literature	GENERIF_SUMMARY	E2F1/IGF-1R Loop Contributes to BRAF Inhibitor Resistance in Melanoma.
Literature	GENERIF_SUMMARY	E2F1 with its surrounding regulatory links identified to play different key roles in the hepatocellular carcinoma (HCC) risk factor dysregulation mechanisms. Hsa-mir-19a showed different effects in the three HCC differential regulation networks and vital regulatory role in hepatitis B virus -related HC and was positively regulated by E2F1 in liver cancer patients with no history of primary risk factors.
Literature	GENERIF_SUMMARY	It uncovered E2F1 and TFDP1 as transport substrates of KPNA2 being retained in the cytoplasm upon KPNA2 ablation, thereby resulting in reduced STMN1 expression.
Literature	GENERIF_SUMMARY	Reciprocal regulation of miR-1205 and E2F1 modulates progression of laryngeal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Studies indicate that patients with prostate cancer with high E2F1 expression exhibit shorter biochemical recurrence-free survival than those with low transcription factor E2F1 (E2F1) expression [Review].
Literature	GENERIF_SUMMARY	MiR-1258 promotes the apoptosis of cervical cancer cells by regulating the E2F1/P53 signaling pathway.
Literature	GENERIF_SUMMARY	LINC00461/miR-4478/E2F1 feedback loop promotes non-small cell lung cancer cell proliferation and migration.
Literature	GENERIF_SUMMARY	Silencing of KIF3A suppressed TNBC cell proliferation by repressing the Rb-E2F signaling pathway and inhibited migration and invasion by repressing epithelial-mesenchymal transition. The tumor size was smaller and the number of lung metastatic nodules was

Main Category	Category	Terms
		lower in KIF3A depletion MDA-MB-231 cell xenograft mice than in the negative control group. In addition, KIF3A overexpression correlated with chemoresistance.
Literature	GENERIF_SUMMARY	A significant role of transcription factors E2F in inflammation and tumorigenesis of nasopharyngeal carcinoma.
Literature	GENERIF_SUMMARY	lncRNA-PLACT1 sustains activation of NF-kappaB pathway through a positive feedback loop with IkappaBalpha/E2F1 axis in pancreatic cancer.
Literature	GENERIF_SUMMARY	Upregulation of miR-34c after silencing E2F transcription factor 1 inhibits paclitaxel combined with cisplatin resistance in gastric cancer cells.
Literature	GENERIF_SUMMARY	The circRNA circSEPT9 mediated by E2F1 and EIF4A3 facilitates the carcinogenesis and development of triple-negative breast cancer.
Literature	GENERIF_SUMMARY	E2F1 mediated DDX11 transcriptional activation promotes hepatocellular carcinoma progression through PI3K/AKT/mTOR pathway.
Literature	GENERIF_SUMMARY	Staufen1 is Essential for Cell-Cycle Transitions and Cell Proliferation Via the Control of E2F1 Expression.
Literature	GENERIF_SUMMARY	The Role of Crosstalk between AR3 and E2F1 in Drug Resistance in Prostate Cancer Cells.
Literature	GENERIF_SUMMARY	E2F1-induced overexpression of SBF2-AS1 promotes the expression of GRB2 by targeting miR-362-3p to facilitate the metastasis of NSCLC
Literature	GENERIF_SUMMARY	The low expression of miR-449a/b and the high expression of E2F1 are both involved in the occurrence and progression of gastric cancer, and miR-449a/b negatively regulates E2F1 to inhibit the proliferation of gastric cancer cells.
Literature	GENERIF_SUMMARY	Redirecting E2F1 to TA-p73 improves cancer therapy through apoptotic induction.
Literature	GENERIF_SUMMARY	Identification of nsSNPs of transcription factor E2F1 predisposing individuals to lung cancer and head and neck cancer.
Literature	GENERIF_SUMMARY	MDM2's dual mRNA binding domains co-ordinate its oncogenic and tumour suppressor activities.
Literature	GENERIF_SUMMARY	p53 inhibits the osteogenic differentiation but does not induce senescence in human dental follicle cells.
Literature	GENERIF_SUMMARY	SNHG12/miR-326/E2F1 feedback loop facilitates the progression of oral squamous cell carcinoma.
Literature	GENERIF_SUMMARY	E2F1 sumoylation as a protective cellular mechanism in oxidative stress response.
Literature	GENERIF_SUMMARY	Silence of ATAD2 inhibits proliferation of colorectal carcinoma via the Rb-E2F1 signaling.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	CircRNA FGFR3 induces epithelial-mesenchymal transition of ovarian cancer by regulating miR-29a-3p/ E2F1 axis.
Literature	GENERIF_SUMMARY	PRMT5 promotes cancer cell migration and invasion through the E2F pathway.
Literature	GENERIF_SUMMARY	A TRAIL-TL1A Paracrine Network Involving Adipocytes, Macrophages, and Lymphocytes Induces Adipose Tissue Dysfunction Downstream of E2F1 in Human Obesity.
Literature	GENERIF_SUMMARY	Activating PIK3CA mutation promotes adipogenesis of adipose-derived stem cells in macrodactyly via up-regulation of E2F1.
Literature	GENERIF_SUMMARY	miR-1258 Regulates Cell Proliferation and Cell Cycle to Inhibit the Progression of Breast Cancer by Targeting E2F1.
Literature	GENERIF_SUMMARY	The oncofetal RNA-binding protein IGF2BP1 is a druggable, post-transcriptional super-enhancer of E2F-driven gene expression in cancer.
Literature	GENERIF_SUMMARY	E2F1-activated SPIN1 promotes tumor growth via a MDM2-p21-E2F1 feedback loop in gastric cancer.
Literature	GENERIF_SUMMARY	MiR-93 suppresses tumorigenesis and enhances chemosensitivity of breast cancer via dual targeting E2F1 and CCND1.
Literature	GENERIF_SUMMARY	SUMOylation of E2F1 Regulates Expression of EZH2.
Literature	GENERIF_SUMMARY	LncRNA-SLC16A1-AS1 induces metabolic reprogramming during Bladder Cancer progression as target and co-activator of E2F1.
Literature	GENERIF_SUMMARY	IL-6-induced acetylation of E2F1 aggravates oxidative damage of retinal pigment epithelial cell line.
Literature	GENERIF_SUMMARY	Downregulation of miR-326 and its host gene beta-arrestin1 induces pro-survival activity of E2F1 and promotes medulloblastoma growth.
Literature	GENERIF_SUMMARY	Long non-coding RNA DLX6-AS1 mediates proliferation, invasion and apoptosis of endometrial cancer cells by recruiting p300/E2F1 in DLX6 promoter region.
Literature	GENERIF_SUMMARY	Oncogenic miR-20b-5p contributes to malignant behaviors of breast cancer stem cells by bidirectionally regulating CCND1 and E2F1.
Literature	GENERIF_SUMMARY	WD Repeat Domain 77 Protein Regulates Translation of E2F1 and E2F3 mRNA.
Literature	GENERIF_SUMMARY	[Advancement of E2F1 in Common Tumors].
Literature	GENERIF_SUMMARY	E2F1-mediated repression of WNT5A expression promotes brain metastasis dependent on the ERK1/2 pathway in EGFR-mutant non-small cell lung cancer.
Literature	GENERIF_SUMMARY	DNMT3A inhibits E2F1-induced arterial marker

Main Category	Category	Terms
		expression and impairs angiogenesis in human umbilical artery endothelial cells.
Literature	GENERIF_SUMMARY	E2F1-induced upregulation of lncRNA HCG18 stimulates proliferation and migration in gastric cancer by binding to miR-197-3p.
Literature	GENERIF_SUMMARY	Human Papillomavirus E7 Oncoprotein Promotes Proliferation and Migration through the Transcription Factor E2F1 in Cervical Cancer Cells.
Literature	GENERIF_SUMMARY	LncRNA-TMPO-AS1 promotes apoptosis of osteosarcoma cells by targeting miR-329 and regulating E2F1.
Literature	GENERIF_SUMMARY	E2F1 Promotes Progression of Bladder Cancer by Modulating RAD54L Involved in Homologous Recombination Repair.
Literature	GENERIF_SUMMARY	Pharmacological Inhibition of CDK8 in Triple-Negative Breast Cancer Cell Line MDA-MB-468 Increases E2F1 Protein, Induces Phosphorylation of STAT3 and Apoptosis.
Literature	GENERIF_SUMMARY	mRNA and miRNA Expression Analyses of the MYC/E2F/miR-17-92 Network in the Most Common Pediatric Brain Tumors.
Literature	GENERIF_SUMMARY	Blockade of deubiquitinating enzyme PSMD14 overcomes chemoresistance in head and neck squamous cell carcinoma by antagonizing E2F1/Akt/SOX2-mediated stemness.
Literature	GENERIF_SUMMARY	The ATPase Pontin is a key cell cycle regulator by amplifying E2F1 transcription response in glioma.
Literature	GENERIF_SUMMARY	Double-targeting CDCA8 and E2F1 inhibits the growth and migration of malignant glioma.
Literature	GENERIF_SUMMARY	MNK1 and MNK2 enforce expression of E2F1, FOXM1, and WEE1 to drive soft tissue sarcoma.
Literature	GENERIF_SUMMARY	Distinct and overlapping roles of ARID3A and ARID3B in regulating E2Fdependent transcription via direct binding to E2F target genes.
Literature	GENERIF_SUMMARY	E2F1 copy number variations in germline and breast cancer
Literature	GENERIF_SUMMARY	E2F1 and E2F2-Mediated Repression of CPT2 Establishes a Lipid-Rich Tumor-Promoting Environment.
Literature	GENERIF_SUMMARY	Long non-coding RNA ZEB1-AS1 promotes proliferation and metastasis of hepatocellular carcinoma cells by targeting miR-299-3p/E2F1 axis.
Literature	GENERIF_SUMMARY	LINC00200 contributes to the chemoresistance to oxaliplatin of gastric cancer cells via regulating E2F1/RAD51 axis.
Literature	GENERIF_SUMMARY	LncRNA SNHG6 promotes G1/S-phase transition in hepatocellular carcinoma by impairing miR-204-5p-

Main Category	Category	Terms
		mediated inhibition of E2F1.
Literature	GENERIF_SUMMARY	LncRNA Lnc-APUE is Repressed by HNF4alpha and Promotes G1/S Phase Transition and Tumor Growth by Regulating MiR-20b/E2F1 Axis.
Literature	GENERIF_SUMMARY	The nucleoplasmic interactions among Lamin A/C-pRB-LAP2alpha-E2F1 are modulated by dexamethasone.
Literature	GENERIF_SUMMARY	E2F1 promotes proliferation and metastasis of clear cell renal cell carcinoma via activation of SREBP1-dependent fatty acid biosynthesis.
Literature	GENERIF_SUMMARY	The E2F1/USP11 positive feedback loop promotes hepatocellular carcinoma metastasis and inhibits autophagy by activating ERK/mTOR pathway.
Literature	GENERIF_SUMMARY	Receptor-Interacting Protein 140 Enhanced Temozolomide-Induced Cellular Apoptosis Through Regulation of E2F1 in Human Glioma Cell Lines.
Literature	GENERIF_SUMMARY	A novel role of miR-326 in colorectal carcinoma by regulating E2F1 expression.
Literature	GENERIF_SUMMARY	Reprogramming of microRNA expression via E2F1 downregulation promotes Salmonella infection both in infected and bystander cells.
Literature	GENERIF_SUMMARY	Long non-coding RNA SNHG16 regulates E2F1 expression by sponging miR-20a-5p and aggravating proliferative diabetic retinopathy.
Literature	GENERIF_SUMMARY	AP-2alpha-Mediated Activation of E2F and EZH2 Drives Melanoma Metastasis.
Literature	GENERIF_SUMMARY	Exosomal miRNA-205 promotes breast cancer chemoresistance and tumorigenesis through E2F1.
Literature	GENERIF_SUMMARY	E2F1 transcriptionally regulates CCNA2 expression to promote triple negative breast cancer tumorigenicity.
Literature	GENERIF_SUMMARY	LINC00319 promotes cancer stem cell-like properties in laryngeal squamous cell carcinoma via E2F1-mediated upregulation of HMGB3.
Literature	GENERIF_SUMMARY	Long non-coding RNA LINC00630 facilitates hepatocellular carcinoma progression through recruiting transcription factor E2F1 to up-regulate cyclin-dependent kinase 2 expression.
Literature	GENERIF_SUMMARY	MicroRNA-205-5p targets E2F1 to promote autophagy and inhibit pulmonary fibrosis in silicosis through impairing SKP2-mediated Beclin1 ubiquitination.
Literature	GENERIF_SUMMARY	FOXO1 inhibits prostate cancer cell proliferation via suppressing E2F1 activated NPRL2 expression.
Literature	GENERIF_SUMMARY	Genetic variations of DNA bindings of FOXA1 and co-factors in breast cancer susceptibility.
Literature	GENERIF_SUMMARY	E2F transcription factor 1 (E2F1) promotes the transforming growth factor TGF-beta1 induced human

Main Category	Category	Terms
		cardiac fibroblasts differentiation through promoting the transcription of CCNE2 gene.
Literature	GENERIF_SUMMARY	CHPF promotes gastric cancer tumorigenesis through the activation of E2F1.
Literature	GENERIF_SUMMARY	Clinical Implication of E2F Transcription Factor 1 in Hepatocellular Carcinoma Tissues.
Literature	GENERIF_SUMMARY	KDM5D inhibits the transcriptional activation of FKBP4 by suppressing the expression of E2F1 in colorectal cancer in males.
Literature	GENERIF_SUMMARY	E2F transcription factor 1 is involved in the phenotypic modulation of esophageal squamous cell carcinoma cells via microRNA-375.
Literature	GENERIF_SUMMARY	A novel function of HRP-3 in regulating cell cycle progression via the HDAC-E2F1-Cyclin E pathway in lung cancer.
Literature	GENERIF_SUMMARY	E2F transcription factor 1 elevates cyclin D1 expression by suppressing transcription of microRNA-107 to augment progression of glioma.
Literature	GENERIF_SUMMARY	CCNE1 and E2F1 Partially Suppress G1 Phase Arrest Caused by Spliceostatin A Treatment.
Literature	GENERIF_SUMMARY	Troponin C-1 Activated by E2F1 Accelerates Gastric Cancer Progression via Regulating TGF-beta/Smad Signaling.
Literature	GENERIF_SUMMARY	Transcription factor p53-mediated activation of miR-519d-3p and downregulation of E2F1 attenuates prostate cancer growth and metastasis.
Literature	GENERIF_SUMMARY	tRNA biogenesis and specific aminoacyl-tRNA synthetases regulate senescence stability under the control of mTOR.
Literature	GENERIF_SUMMARY	E2F transcription factor 1 (E2F1) enhances the proliferation, invasion and EMT of trophoblast cells by binding to Zinc Finger E-Box Binding Homeobox 1 (ZEB1).
Literature	GENERIF_SUMMARY	E2F and STAT3 provide transcriptional synergy for histone variant H2AZ activation to sustain glioblastoma chromatin accessibility and tumorigenicity.
Literature	GENERIF_SUMMARY	N6-methyladenosine-induced SVIL antisense RNA 1 restrains lung adenocarcinoma cell proliferation by destabilizing E2F1.
Literature	GENERIF_SUMMARY	LncRNA HOXA10-AS Activated by E2F1 Facilitates Proliferation and Migration of Nasopharyngeal Carcinoma Cells Through Sponging miR-582-3p to Upregulate RAB31.
Literature	GENERIF_SUMMARY	Role of Lysine-Specific Demethylase 1 in Metabolically Integrating Osteoclast Differentiation and Inflammatory Bone Resorption Through Hypoxia-Inducible Factor 1alpha and E2F1.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	NCAPD3 promotes prostate cancer progression by up-regulating EZH2 and MALAT1 through STAT3 and E2F1.
Literature	GENERIF_SUMMARY	Actinidia chinensis Planch Root extract suppresses the growth and metastasis of hypopharyngeal carcinoma by inhibiting E2F Transcription Factor 1-mediated MNX1 antisense RNA 1.
Literature	GENERIF_SUMMARY	LncRNA OSER1-AS1 regulates the inflammation and apoptosis of rheumatoid arthritis fibroblast like synoviocytes via regulating miR-1298-5p/E2F1 axis.
Literature	GENERIF_SUMMARY	E2F1-induced lncRNA, EMSLR regulates lncRNA LncPRESS1.
Literature	GENERIF_SUMMARY	E2F1-mediated AUF1 upregulation promotes HCC development and enhances drug resistance via stabilization of AKR1B10.
Literature	GENERIF_SUMMARY	LOC100130075 Promotes Cervical Cancer Progression by Activating MDM2 Transcription through E2F1.
Literature	GENERIF_SUMMARY	An Insight into the Role of E2F1 in Breast Cancer Progression, Drug Resistance, and Metastasis.
Literature	GENERIF_SUMMARY	Ribosomal protein L5 (RPL5)/ E2F transcription factor 1 (E2F1) signaling suppresses breast cancer progression via regulating endoplasmic reticulum stress and autophagy.
Literature	GENERIF_SUMMARY	Positive regulation of ataxia-telangiectasia-mutated protein (ATM) by E2F transcription Factor 1 (E2F-1) in cisplatin-resistant nasopharyngeal carcinoma cells.
Literature	GENERIF_SUMMARY	circ0125803 facilitates tumor progression by sponging miR-197-5p and upregulating E2F1 in neuroblastoma.
Literature	GENERIF_SUMMARY	NSD2 activates the E2F transcription factor 1/Y-box binding protein 2 axis to promote the malignant development of oral squamous cell carcinoma.
Literature	GENERIF_SUMMARY	The human ion channel TRPM2 modulates cell survival in neuroblastoma through E2F1 and FOXM1.
Literature	GENERIF_SUMMARY	The miR-532-E2F1 feedback loop contributes to gastric cancer progression.
Literature	GENERIF_SUMMARY	Nuclear respiratory factor 1 promotes the growth of liver hepatocellular carcinoma cells via E2F1 transcriptional activation.
Literature	GENERIF_SUMMARY	E2F1-activated LINC01224 drives esophageal squamous cell carcinoma cell malignant behaviors via targeting miR-6884-5p/DVL3 axis and activating Wnt/ beta-catenin signaling pathway.
Literature	GENERIF_SUMMARY	E2F1 promotes Warburg effect and cancer progression via upregulating ENO2 expression in Ewing sarcoma.
Literature	GENERIF_SUMMARY	Prognostic analysis of E2F transcription factors E2F1 and E2F3 in four independent pediatric neuroblastoma cohorts.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Genetic Polymorphism of E2F1 Influences Susceptibility to Ovarian Cancer in a Chinese Population.
Literature	GENERIF_SUMMARY	E2F Transcription Factor 1 Activates FKBP Prolyl Isomerase 4 to Promote Angiogenesis in Cervical Squamous Cell Carcinoma Via the PI3K/AKT Signaling Pathway.
Literature	GENERIF_SUMMARY	The transcription factor E2F1 controls the GLP-1 receptor pathway in pancreatic beta cells.
Literature	GENERIF_SUMMARY	E2F1 induced neuroblastoma cell migration and invasion via activation of CENPE/FOXM1 signaling pathway.
Literature	GENERIF_SUMMARY	E2F1 transcription factor mediates a link between fat and islets to promote beta cell proliferation in response to acute insulin resistance.
Literature	GENERIF_SUMMARY	E2F1 promotes cell cycle progression by stabilizing spindle fiber in colorectal cancer cells.
Literature	GENERIF_SUMMARY	E2F1-initiated transcription of PRSS22 promotes breast cancer metastasis by cleaving ANXA1 and activating FPR2/ERK signaling pathway.
Literature	GENERIF_SUMMARY	PLAGL2 promotes the stemness and is upregulated by transcription factor E2F1 in human lung cancer.
Literature	GENERIF_SUMMARY	E2F1 combined with LINC01004 super-enhancer to promote hepatocellular carcinoma cell proliferation and metastasis.
Literature	GENERIF_SUMMARY	hMSH2 coordinated with the expression of E2F1 promotes platinum response in epithelial ovarian cancer.
Literature	GENERIF_SUMMARY	PRB inhibited cell proliferation through let-7b-E2F1 in breast cancer.
Literature	GENERIF_SUMMARY	ANKRD22 promotes glioma proliferation, migration, invasion, and epithelial-mesenchymal transition by upregulating E2F1-mediated MELK expression.
Literature	GENERIF_SUMMARY	DTL is a Novel Downstream Gene of E2F1 that Promotes the Progression of Hepatocellular Carcinoma.
Literature	GENERIF_SUMMARY	MCM6 promotes intrahepatic cholangiocarcinoma progression by upregulating E2F1 and enhancing epithelial-mesenchymal transition.
Literature	GENERIF_SUMMARY	beta-Cell-Specific E2f1 Deficiency Impairs Glucose Homeostasis, beta-Cell Identity, and Insulin Secretion.
Literature	GENERIF_SUMMARY	Prognostic role of E2F1 gene expression in human cancer
Literature	GENERIF_SUMMARY	Identification of the E2F1-RAD51AP1 axis as a key factor in MGMT-methylated GBM TMZ resistance.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Clinical evidence for a role of E2F1-induced replication stress in modulating tumor mutational burden and immune microenvironment.
Literature	GENERIF_SUMMARY	E2F1 rs3213150 polymorphism influences cytarabine sensitivity and prognosis in patients with acute myeloid leukemia.
Literature	GENERIF_SUMMARY	E2F1 Mediates SOX17 Deficiency-Induced Pulmonary Hypertension.
Literature	GENERIF_SUMMARY	miRNA-329-3p suppresses proliferation and metastasis of endometrial carcinoma through downregulating E2F1.
Literature	GENERIF_SUMMARY	The E2F1/MELTF axis fosters the progression of lung adenocarcinoma by regulating the Notch signaling pathway.
Literature	GENERIF_SUMMARY	Transcription Factor E2F1 Enhances Hepatocellular Carcinoma Cell Proliferation and Stemness by Activating GINS1.
Literature	GENERIF_SUMMARY	NEIL3 promotes cell proliferation of ccRCC via the cyclin D1-Rb-E2F1 feedback loop regulation.
Literature	GENERIF_SUMMARY	DLGAP5 triggers proliferation and metastasis of bladder cancer by stabilizing E2F1 via USP11.
Literature	GENERIF_SUMMARY	Analysis of E2F1 single-nucleotide polymorphisms reveals deleterious non-synonymous substitutions that disrupt E2F1-RB protein interaction in cancer.
Literature	GENERIF_SUMMARY	The association of E2F1 and E2F2 single nucleotide polymorphisms with laryngeal squamous cell carcinoma pathomorphological features.
Literature	GENERIF_SUMMARY	Hsa_circ_0007990 promotes breast cancer growth via inhibiting YBX1 protein degradation to activate E2F1 transcription.
Literature	GENERIF_SUMMARY	E2F1 Mediates Traumatic Brain Injury and Regulates BDNF-AS to Promote the Progression of Alzheimer's Disease.
Literature	GENERIF_SUMMARY	RNA m[5]C modification upregulates E2F1 expression in a manner dependent on YBX1 phase separation and promotes tumor progression in ovarian cancer.
Literature	GENERIF_SUMMARY	New link between RNH1 and E2F1
Literature	GENERIF_SUMMARY	CircMYBL2 facilitates hepatocellular carcinoma progression by regulating E2F1 expression.
Literature	GENERIF_SUMMARY	CDCA5 accelerates progression of breast cancer by promoting the binding of E2F1 and FOXM1.
Literature	GENERIF_SUMMARY	Epigenetically associated IGF2BP3 upregulation promotes cell proliferation by regulating E2F1 expression in hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	DNMT1-Mediated the Downregulation of FOXF1

Main Category	Category	Terms
		Promotes High Glucose-induced Podocyte Damage by Regulating the miR-342-3p/E2F1 Axis.
Literature	GENERIF_SUMMARY	ARHGEF39 targeted by E2F1 fosters hepatocellular carcinoma metastasis by mediating fatty acid metabolism.
Literature	GENERIF_SUMMARY	LncRNA HAGLR regulates gastric cancer progression by regulating the miR-20a-5p/E2F1 axis.
Pathways	KEGG_PATHWAY	Endocrine resistance
Pathways	KEGG_PATHWAY	Cell cycle
Pathways	KEGG_PATHWAY	Mitophagy - animal
Pathways	KEGG_PATHWAY	Cellular senescence
Pathways	KEGG_PATHWAY	Cushing syndrome
Pathways	KEGG_PATHWAY	Hepatitis C
Pathways	KEGG_PATHWAY	Hepatitis B
Pathways	KEGG_PATHWAY	Human cytomegalovirus infection
Pathways	KEGG_PATHWAY	Human papillomavirus infection
Pathways	KEGG_PATHWAY	Human T-cell leukemia virus 1 infection
Pathways	KEGG_PATHWAY	Kaposi sarcoma-associated herpesvirus infection
Pathways	KEGG_PATHWAY	Epstein-Barr virus infection
Pathways	KEGG_PATHWAY	Pathways in cancer
Pathways	KEGG_PATHWAY	MicroRNAs in cancer
Pathways	KEGG_PATHWAY	Chemical carcinogenesis - receptor activation
Pathways	KEGG_PATHWAY	Pancreatic cancer
Pathways	KEGG_PATHWAY	Glioma
Pathways	KEGG_PATHWAY	Prostate cancer
Pathways	KEGG_PATHWAY	Melanoma
Pathways	KEGG_PATHWAY	Bladder cancer
Pathways	KEGG_PATHWAY	Chronic myeloid leukemia
Pathways	KEGG_PATHWAY	Small cell lung cancer
Pathways	KEGG_PATHWAY	Non-small cell lung cancer
Pathways	KEGG_PATHWAY	Breast cancer
Pathways	KEGG_PATHWAY	Hepatocellular carcinoma
Pathways	KEGG_PATHWAY	Gastric cancer
Pathways	REACTOME_PATHWAY	Apoptosis

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	Intrinsic Pathway for Apoptosis
Pathways	REACTOME_PATHWAY	Activation of NOXA and translocation to mitochondria
Pathways	REACTOME_PATHWAY	Inhibition of replication initiation of damaged DNA by RB1/E2F1
Pathways	REACTOME_PATHWAY	E2F mediated regulation of DNA replication
Pathways	REACTOME_PATHWAY	Activation of BH3-only proteins
Pathways	REACTOME_PATHWAY	Developmental Biology
Pathways	REACTOME_PATHWAY	Transcription of E2F targets under negative control by DREAM complex
Pathways	REACTOME_PATHWAY	Transcription of E2F targets under negative control by p107 (RBL1) and p130 (RBL2) in complex with HDAC1
Pathways	REACTOME_PATHWAY	Activation of PUMA and translocation to mitochondria
Pathways	REACTOME_PATHWAY	G0 and Early G1
Pathways	REACTOME_PATHWAY	Signaling by NOTCH
Pathways	REACTOME_PATHWAY	Signal Transduction
Pathways	REACTOME_PATHWAY	Cell Cycle
Pathways	REACTOME_PATHWAY	Disease
Pathways	REACTOME_PATHWAY	Pre-NOTCH Transcription and Translation
Pathways	REACTOME_PATHWAY	Pre-NOTCH Expression and Processing
Pathways	REACTOME_PATHWAY	Generic Transcription Pathway
Pathways	REACTOME_PATHWAY	Cellular responses to stress
Pathways	REACTOME_PATHWAY	Oxidative Stress Induced Senescence
Pathways	REACTOME_PATHWAY	Cellular Senescence
Pathways	REACTOME_PATHWAY	Oncogene Induced Senescence
Pathways	REACTOME_PATHWAY	Transcriptional Regulation by TP53
Pathways	REACTOME_PATHWAY	Mitotic G2-G2/M phases
Pathways	REACTOME_PATHWAY	Mitotic G1 phase and G1/S transition
Pathways	REACTOME_PATHWAY	Programmed Cell Death
Pathways	REACTOME_PATHWAY	TP53 Regulates Transcription of Cell Cycle Genes
Pathways	REACTOME_PATHWAY	TP53 Regulates Transcription of Genes Involved in G1 Cell Cycle Arrest
Pathways	REACTOME_PATHWAY	G2 Phase
Pathways	REACTOME_PATHWAY	Cyclin E associated events during G1/S transition
Pathways	REACTOME_PATHWAY	G1/S-Specific Transcription

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	G1/S Transition
Pathways	REACTOME_PATHWAY	Cyclin D associated events in G1
Pathways	REACTOME_PATHWAY	G1 Phase
Pathways	REACTOME_PATHWAY	S Phase
Pathways	REACTOME_PATHWAY	Cell Cycle, Mitotic
Pathways	REACTOME_PATHWAY	Cyclin A:Cdk2-associated events at S phase entry
Pathways	REACTOME_PATHWAY	RNA Polymerase II Transcription
Pathways	REACTOME_PATHWAY	Gene expression (Transcription)
Pathways	REACTOME_PATHWAY	Transcriptional Regulation by E2F6
Pathways	REACTOME_PATHWAY	Cellular responses to stimuli
Pathways	REACTOME_PATHWAY	Transcriptional regulation of granulopoiesis
Pathways	REACTOME_PATHWAY	Aberrant regulation of mitotic G1/S transition in cancer due to RB1 defects
Pathways	REACTOME_PATHWAY	Defective binding of RB1 mutants to E2F1,(E2F2, E2F3)
Pathways	REACTOME_PATHWAY	Diseases of mitotic cell cycle
Pathways	REACTOME_PATHWAY	Aberrant regulation of mitotic cell cycle due to RB1 defects
Pathways	WIKIPATHWAYS	miRNA regulation of DNA damage response
Pathways	WIKIPATHWAYS	miRNAs involved in DNA damage response
Pathways	WIKIPATHWAYS	Cell cycle
Pathways	WIKIPATHWAYS	Integrated cancer pathway
Pathways	WIKIPATHWAYS	Integrated breast cancer pathway
Pathways	WIKIPATHWAYS	miR targeted genes in lymphocytes
Pathways	WIKIPATHWAYS	miR targeted genes in muscle cell
Pathways	WIKIPATHWAYS	Glioblastoma signaling
Pathways	WIKIPATHWAYS	Androgen receptor network in prostate cancer
Pathways	WIKIPATHWAYS	Adipogenesis
Pathways	WIKIPATHWAYS	Spinal cord injury
Pathways	WIKIPATHWAYS	Retinoblastoma gene in cancer
Pathways	WIKIPATHWAYS	Aryl hydrocarbon receptor pathway
Pathways	WIKIPATHWAYS	Mammary gland development involution stage 4 of 4
Pathways	WIKIPATHWAYS	Bladder cancer
Pathways	WIKIPATHWAYS	MECP2 and associated Rett syndrome

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	DNA IR double strand breaks and cellular response via ATM
Pathways	WIKIPATHWAYS	H19 Rb E2F1 and CDK beta catenin in colorectal cancer
Pathways	WIKIPATHWAYS	Prader Willi and Angelman syndrome
Pathways	WIKIPATHWAYS	DNA IR damage and cellular response via ATR
Pathways	WIKIPATHWAYS	Non small cell lung cancer
Pathways	WIKIPATHWAYS	Breast cancer pathway
Pathways	WIKIPATHWAYS	Pancreatic adenocarcinoma pathway
Pathways	WIKIPATHWAYS	Effect of progerin on genes involved in progeria
Pathways	WIKIPATHWAYS	EGF EGFR signaling
Pathways	WIKIPATHWAYS	G1 to S cell cycle control
Pathways	WIKIPATHWAYS	Small cell lung cancer
Pathways	WIKIPATHWAYS	Head and neck squamous cell carcinoma
Pathways	WIKIPATHWAYS	Melanoma
Pathways	WIKIPATHWAYS	Overlap between signal transduction pathways contributing to LMNA laminopathies
Pathways	WIKIPATHWAYS	Pleural mesothelioma
Pathways	WIKIPATHWAYS	Sphingolipid metabolism in senescence
Pathways	WIKIPATHWAYS	Cancer pathways
Pathways	WIKIPATHWAYS	Transcriptional regulation of memory B cell differentiation
Pathways	WIKIPATHWAYS	Senescence and autophagy in cancer
Pathways	WIKIPATHWAYS	DNA damage response
Protein Domains	INTERPRO	E2F_WHTH_DNA-bd_dom
Protein Domains	INTERPRO	E2F
Protein Domains	INTERPRO	E2F_CC-MB
Protein Domains	INTERPRO	WH-like_DNA-bd_sf
Protein Domains	INTERPRO	WH_DNA-bd_sf
Protein Domains	INTERPRO	E2F-DP_heterodim
Protein Domains	SMART	E2F_TDP